TaxonVision-MLOps: Enterprise Strategy & Architecture

High-Throughput Biodiversity Computer Vision, Mathematical Uncertainty Guarantees & Bare-Metal MLOps

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Production Architecture Presentation

Curriculum Overview

Four-Pillar Architectural Taxonomy

Part I: Domain, Hardware & Repo Topology

Ch 01: Problem & Operational Maturity
Ch 02: Hardware Architecture & Infrastructure
Ch 03: Repository Topology & Configurations

Part II: MLOps Stack, DVC DAG & Code Architecture

Ch 04: MLOps Stack, Databases & Storage
Ch 05: Epistemic Diamond DVC DAG & DagsHub
Ch 06: Software Architecture & Subpackages

Part III: Fast-Path Serving, Data & Model Engineering

Ch 07: Pipeline Architecture & Fast-Path Serving
Ch 08: Data Engineering & Storage (Parquet/DVC)
Ch 09: Model Engineering & Bayesian Tuning

Part IV: Epistemic Safety, CI/CD, Observability & Vision-RAG

Ch 10: Epistemic Safety & Conformal Bounds
Ch 11: CI/CD, Containers & Kubernetes (k3s)
Ch 12: Observability, Drift & Active Learning
Ch 13: Delivery Roadmap & Core Principles
Ch 14: Multimodal Knowledge Graph & Vision-RAG

TaxonVision-MLOps Architecture

CHAPTER 01

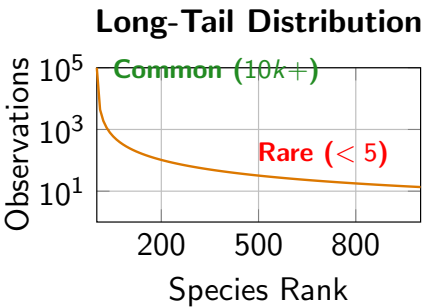
Problem & Operational Maturity

Extreme Fine-Grained Biology, Toy ML Pitfalls & MLOps Level 4

Status: **[Implemented & Verified]**

Biological Domain Challenges

- **Long-Tail Distributions:** Unlike Kaggle datasets (e.g., CIFAR-10), biodiversity data is severely imbalanced. Common species have 10,000+ images, while rare ones have < 5.
- **Fine-Grained Taxonomy:** Inter-class variance can be microscopic (e.g., insect wing venation, fungal spore color).
- **High Intra-Class Variance:** Extreme visual changes due to life stages, sexual dimorphism, lighting, and foliage occlusion.



The "50 TB Trap": Architectural Failure Modes

- **Naive Approach:** Attempting to download massive biological datasets (e.g., GBIF, iNaturalist) directly to local NVMe drives.
- **Consequences:** Saturates workstation storage instantly, introduces massive IO latency (uncompressing millions of JPEGs), and breaks reproducibility.
- **Production Solution (Zero-Copy Streaming):**
  - Stream data directly from S3-compatible object storage.
  - Never download raw high-res images locally for training.
  - Decouple feature extraction (vision backbones) from model experimentation. Cache high-dimensional embeddings instead of raw pixels.

Open Science & Legal Constraints

- **The IP Liability Problem:** Training on randomly scraped web images creates severe legal toxicity, risking model deletion orders.
- **DarwinCore Audit Gates:** TaxonVision implements strict, automated CI gates inspecting DarwinCore metadata before ingestion.
- **Allowed Licenses:**
  - **CC0:** Public Domain (Unrestricted)
  - **CC-BY:** Attribution Required (Manifests generated automatically)
  - **CC-BY-NC:** Non-Commercial (Permitted for this academic scope)
- **Zero-Trust Policy:** Unlicensed data is rejected by default.

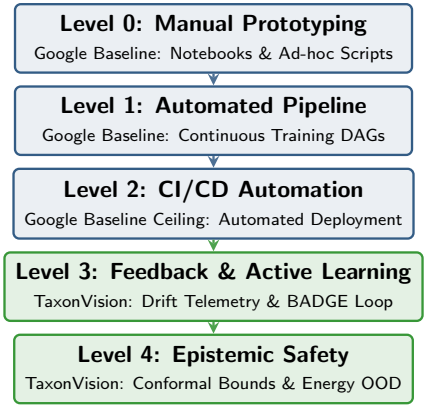
Operational Maturity: 5-Tier Framework (Adapted from Google MLOps)

Extending Standard Google MLOps Levels 0–2 with Epistemic Safety Tiers

5-Tier Operational Maturity Spectrum

- **Level 0 (Manual):** Notebooks, unversioned state, manual handoffs *(Google Baseline)*
- **Level 1 (Pipeline):** Automated ML DAGs, continuous training baseline *(Google Baseline)*
- **Level 2 (CI/CD):** CI/CD pipelines, dynamic ONNX export *(Google Baseline Ceiling)*
- **Level 3 (Feedback):** Concept drift telemetry (W_1 / MMD) & BADGE active learning *(TaxonVision)*
- **Level 4 (Epistemic Safety):** Split Conformal ($1 - \alpha \geq 0.95$) & Energy OOD rejection *(TaxonVision)*

Status: [Partially Implemented / In Progress]



CHAPTER 02

Hardware Architecture & Infrastructure

Physical Topography, RTX 5070 Ti Time-Slicing & Thermal Budgets

Status: [Implemented & Verified]

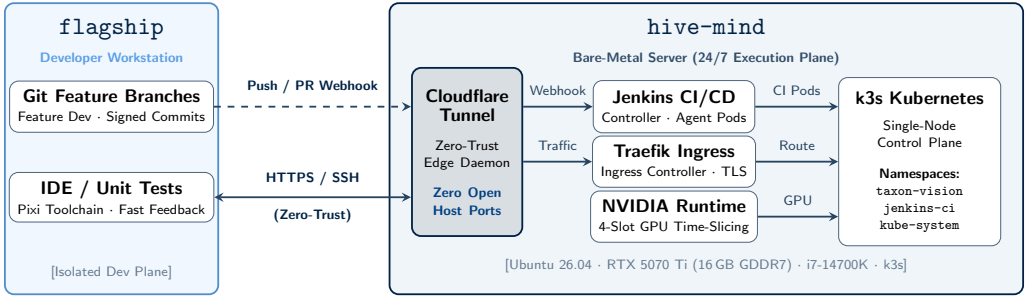
Architectural Philosophy

Instead of relying on expensive public cloud instances, TaxonVision uses an on-premise dual-node topology to ensure continuous availability, high performance, and zero bandwidth egress fees.

- **Developer Workstation:** Dedicated to fast iteration loops, writing code, testing micro-pipelines, and running local unit tests.
- **Bare-Metal Server:** Operates 24/7 hosting the core infrastructure.
- **Decoupled Environments:** Prevents developer experiments from starving production inference of GPU memory or crashing background training jobs.

Infrastructure Topology

Dual-Node Diagram



Compute Hardware Constraints

Our inference engine is powered by the NVIDIA GeForce RTX 5070 Ti, built on the SM 120 Blackwell architecture, equipped with 16 GB of GDDR7 VRAM.

- **Compute Throughput:** Leverages 4th-generation Tensor Cores optimized for INT8, FP16, and BF16 matrix multiplication.
- **Strict Memory Boundaries:** Unlike data center H100s (80 GB HBM3), the 16 GB hard limit demands aggressive memory budgeting and zero dynamic allocation.
- **Frozen Backbones:** Requires pre-compiled backbones and robust ONNX optimizations to prevent CUDA Out of Memory (OOM) errors.

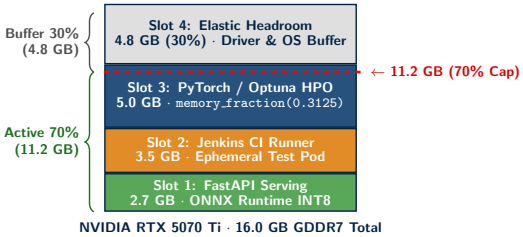
Resource Virtualization Strategy

Allocating an entire GPU to a single API pod is highly inefficient. We use Kubernetes GPU time-slicing to share the RTX 5070 Ti across multiple workloads.

- **4-Slot Time-Slicing:** The physical GPU is virtualized into 4 slots via the NVIDIA Container Toolkit.
- **The 70% Invariant:** A strict 11.2 GB (70% of 16 GB) VRAM ceiling is enforced per container to prevent memory fragmentation and cascading kernel panics.
- **Concurrency:** Enables active learning inference, feature extraction, background testing, and the web API to run safely on the same hardware.

GPU Memory Allocation

Time-Sliced VRAM Partitioning & Process Capping



Resource Virtualization & Workload Budgets

- **4 Virtual Slices:** Declared in `nvidia-time-slicing-config.yaml`.
- **PyTorch / Optuna (Slot 3):** Confined to 5.0 GB physical VRAM via `memory_fraction(0.3125)`.
- **FastAPI Serving (Slot 1):** Zero-allocation INT8 ONNX Runtime engine requires < 2.7 GB.

The 70% OOM Isolation Invariant

- **11.2 GB Active Ceiling:** Cumulative active pod memory cannot exceed 70%, eliminating cascading OOM panics.
- **4.8 GB (30%) Buffer Margin:** Reserved as elastic headroom for OS displays and CUDA drivers.

Secure Ingress Topology

Exposing bare-metal clusters to the public internet via router port-forwarding invites brute-force and DDoS attacks. We leverage zero-open-ports network security.

- **Outbound Encrypted Tunnels:** A persistent `cloudflared` daemon inside the cluster creates outbound connections to Cloudflare's edge network.
- **Zero Exposed Host Ports:** The router blocks all inbound traffic. Traffic routing and TLS termination happen internally via Traefik.
- **Identity Verification:** Operator traffic is authenticated at the Cloudflare edge before ever reaching the on-premise infrastructure.

TaxonVision-MLOps Architecture

CHAPTER 03

Repository Architecture & Configurations

Monorepo Topology, Unified Parameters & Agentic Governance

Status: **[Implemented & Verified]**

Repository Architecture

Directory Tree Topology: Core Governance & Infrastructure

Governance & Workspace Domains

- `.agents/`: AI agent persona routing, 12 specialized roles, 7 behavioral rules, and 6 executable workflows.
- `.dvc/`: Content-addressable dataset cache configurations, local SeaweedFS S3 endpoints, and DagsHub remotes.
- `.github/workflows/`: Declarative CI/CD pipeline (`ci.yml`) with multi-pass pre-commit, coverage, and OKF gates.
- `config/`: Declarative YAML parameters (`params.yaml`, `default_config.yaml`, `taxa_catalog.yaml`).

Data, Deployment & Documentation

- `data/`: Versioned Parquet manifests (`data/manifests/`) and 4-way stratified embedding tensors (`data/embeddings/`).
- `deploy/`: Production Dockerfile (`deploy/docker/`), local Compose stack (`deploy/compose/`), and k3s manifests (`deploy/k8s/`).
- `docs/`: Zensical MkDocs markdown specifications and complete LaTeX Beamer/Article source trees.
- `openwiki/`: Open Knowledge Format (OKF v0.2) concept documents and cryptographic evidence claim graphs.

Repository Architecture

Directory Tree Topology: Application Source, Tests & Scripts

Application Core (`src/taxon_vision/`)

- `config/`: Typed Pydantic schemas & YAML loader.
- `data/`: S3 streaming, iNaturalist client, license gates.
- `models/`: Backbone factory, Class-Balanced loss (CBL).
- `inference/`: ONNX Runtime engine, head projector, Grad-CAM.
- `uncertainty/`: Split Conformal predictor & OOD energy.
- `service/`: FastAPI routes, Jinja2/HTMX dashboard.
- `pipelines/`: 7 canonical DVC stage CLI entrypoints.
- `monitoring/`: Prometheus telemetry and drift detectors.

Quality, Automation & Reports

- `tests/unit/`: Granular component tests for configs, pipelines, models, quantizers, and conformal engines.
- `tests/integration/`: Full FastAPI endpoint simulations and HTMX interaction sweeps.
- `tests/invariants/`: Mathematical coverage tests and pure Pixi toolchain assertions.
- `scripts/`: Automation utilities grouped into `ci/`, `data/`, and `governance/`.
- `reports/`: Machine-readable JSON metrics for training, license audits, and Pareto benchmarks.

Root Governance

Build Specifications, Dependency Lockfiles & Task Runners

Hermetic Build Specifications

- `pyproject.toml`: Declares multi-target Pixi environments (`default`, `dev`, `ci-dev`) across `conda-forge`, `pytorch`, and `nvidia` channels. Configures Ruff, Mypy, and Pytest.
- `pixi.lock`: Cryptographically pinned binary lockfile guaranteeing bit-for-bit reproducible environments across workstations, CI nodes, and Kubernetes pods.
- `zensical.toml`: Configuration for the Zensical MkDocs documentation compiler with KaTeX and Mermaid.

Orchestration & Pipeline Graphs

- `Justfile`: Polyglot task runner automating developer workflows (`just check`, `just test`, `just repro`, `just optuna-k8s`, `just audit-license`).
- `dvc.yaml`: Declarative 7-stage Epistemic Diamond DAG defining explicit parameter, code, and data dependencies.
- `dvc.lock`: Immutable state tracking content hashes (SHA-256 / MD5) for all intermediate tensors and ONNX binaries.

Parameter Encyclopedia

`params.yaml`: Data Ingestion & Vision Backbone Architecture

Parameter Key	Default	Consuming Stage	Engineering & Mathematical Purpose
`ingestion.taxa_to_fetch`	10	`ingest`, `audit`	Number of target species catalog entries queried from iNaturalist.
`ingestion.per_page`	30	`ingest`	Observation batch size per API page during catalog ingestion.
`model.extractor`	`dinov3`	`extract_features`, `train`, `export`	Foundation vision backbone (`dinov3`, `bioclip2`, `dinov2`, `mobilenetv4`).
`model.image_size`	224	`extract_features`, `export`	Spatial input resolution ($224 \times 224 \times 3$) for backbone resizing.
`model.num_classes`	10	`train`, `export`	Dimension of linear classification head output logit vector.

Architecture Invariant: Parameter Centrality

Parameters defined in `params.yaml` drive the DVC execution graph; changing any key automatically invalidates downstream stages and forces a deterministic recalculation of affected stages.

Parameter Encyclopedia

`params.yaml`: Feature Extraction & Head Training Parameters

Parameter Key	Default	Consuming Stage	Engineering & Mathematical Purpose
`feature_extraction.batch_size`	64	`extract_features`	Batch size for batched forward passes through frozen backbones.
`feature_extraction.val_split`	0.15	`extract_features`	Stratified validation split ratio for early stopping & tuning.
`feature_extraction.cal_split`	0.15	`extract_features`	Independent calibration split for Split Conformal Prediction.
`feature_extraction.test_split`	0.10	`extract_features`	Holdout Pareto benchmark test partition (60% train remainder).
`training.epochs`	5	`train`	Epoch count for training linear classification heads on cached tensors.
`training.learning_rate`	0.001	`train`	Initial learning rate for AdamW optimization.
`training.cb_beta`	0.999	`train`	Effective sample weighting hyperparameter β for Class-Balanced loss.
`training.tune`	false	`train`	Toggle for Optuna Bayesian hyperparameter optimization sweeps.

Parameter Encyclopedia

`params.yaml`: Export, Conformal Bounds & Pareto Evaluation

Parameter Key	Default	Consuming Stage	Engineering & Mathematical Purpose
`export.quantize_int8`	`true`	`export_onnx`	Enables ONNX dynamic INT8 quantization for $4\times$ smaller model weight size.
`export.opset_version`	17	`export_onnx`	ONNX specification operator set version for export compatibility.
`conformal.alpha`	0.05	`calibrate`	Target error rate, guaranteeing $1-\alpha = 95\%$ statistical coverage.
`conformal.temperature`	1.0	`calibrate`	Post-hoc logit temperature scaling parameter prior to conformal scoring.
`evaluation.benchmark_runs`	100	`evaluate`	Iteration count for p95 execution latency profiling under ONNX Runtime.
`evaluation.p95_budget_ms`	15.0	`evaluate`	Maximum allowable p95 inference latency threshold for production gating.

Zero-Allocation Latency Gate

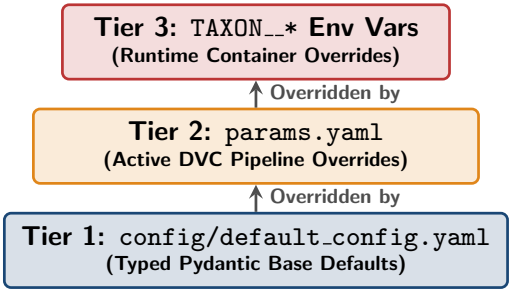
The evaluate stage verifies that the quantized INT8 model satisfies both the 95% coverage guarantee and the 15 ms latency budget before marking the pipeline as green.

Configuration Engine

config/default_config.yaml: Centralized Settings Engine

Declarative Service & Storage Defaults

- **Service Binding:** `service.host: "0.0.0.0"`, `service.port: 8000`, `service.workers: 4`.
- **Filesystem Layout:** `data_dir: "data"`, `artifacts_dir: "models/checkpoints"`, `reports_dir: "reports"`.
- **MLflow Registry:** `mlflow.tracking_uri: "https://dagshub.com/.../mlflow"`.
- **Active Learning:** `active_learning.uncertainty_threshold: 0.75`.



- **Precedence Rule:** Tier 3 $\succ$ Tier 2 $\succ$ Tier 1.
- **Zero Magic Numbers:** Statically validated by Pydantic schemas in `config.py`.

Specialized Manifests

Domain Catalogs, Conformal Calibration & Prometheus Scrapes

Calibration & Catalog Declarations

- `config/conformal_config.yaml`: Serializes empirical calibration outputs:
  - Quantile threshold: $\hat{q} = 0.8405$
  - OOD energy gate: $\tau_{\text{ood}} = -2.2792$
  - Nominal coverage: $1 - \alpha = 0.95$
  - Calibrated sample size: $N_{\text{cal}} = 169$
- `config/taxa_catalog.yaml`: Authoritative DarwinCore definitions for target species, resolving GBIF taxonomy and class indices.

Telemetry & Formatting Governance

- `config/prometheus.yaml`: Defines Prometheus scrape daemon targets, 15 s intervals, and metric paths (`/metrics`) for the API.
- `config/lint/.markdownlint.yaml`: Strict Markdown style rules (standard hyphens, 4-space indented asterisks, blank lines surrounding fences).

Automation Tooling

scripts/: Continuous Integration, Data Ingestion & Governance

scripts/ci/

- `local_ci.sh`: Native host reproduction of GitHub Actions CI sweeps without Docker overhead.
- `jenkins_gpu_train.sh`: Modular script executed in the Jenkins k3s GPU agent pod.
- `verify_deployment.sh`: Post-deploy smoke test asserting pod readiness and metrics.

scripts/data/

- `download_samples.py`: Fetches verified open-license sample photographs for catalog taxa from iNaturalist S3.

Compatibility Shims:

- `scripts/validate_okf.py`
- `scripts/visualize_okf.py`

scripts/governance/

- `validate_okf.py`: Google Open Knowledge Format linter verifying frontmatter and links.
- `visualize_okf.py`: Standalone D3/SVG knowledge graph compiler (`docs/viz.html`).
- `verify_markdown_rules.py`: Enforces Markdown Rule 04 formatting.

Deployment Topology

deploy/: Containers, Compose Stacks & Kubernetes Manifests

Containerization Architecture

- deploy/docker/Dockerfile: Multi-stage production container based on Pixi. Strips build tooling, runs as non-root user taxon, and installs ONNX Runtime CPU.
- deploy/compose/ docker-compose.local.yaml: Complete standalone local deployment running FastAPI, SeaweedFS S3, Prometheus, and Grafana.

Kubernetes Cluster Manifests (deploy/k8s/)

- api-deployment.yaml: Rolling-update Deployment, ConfigMap, ClusterIP Service, and HTTP probes.
- jenkins-gpu-pod.yaml: Ephemeral GPU runner pod requesting nvidia.com/gpu: "1" and host caches.
- jenkins-agent-rbac.yaml: Least-privilege ServiceAccount and RoleBinding for CI runners.
- mlops-storage-and-optuna.yaml: PVC for SQLite and Optuna web dashboard.
- nvidia-time-slicing-config.yaml: 4-way time-slicing plugin config for RTX 5070 Ti.

Agentic Governance

`.agents/roles/`: 12 Path-Scoped AI Roles & Responsibilities

Core ML & Pipeline Roles

- **@orchestrator**: PM, milestone coordinator, OKF guardian.
- **@taxonomic-architect**: DarwinCore schemas, GBIF tree mappings.
- **@mlops-data-engineer**: S3 streaming, DVC hashes, licensing.
- **@model-engineer**: Timm factory, Class-Balanced loss, training.
- **@inference-optimizer**: ONNX export, INT8 quantization, CAM.
- **@uncertainty-safety-engineer**: Conformal prediction, energy OOD.

Platform, Safety & Ops Roles

- **@active-learning-engineer**: BADGE, CoreSet, human triage queue.
- **@observability-engineer**: Prometheus metrics, drift telemetry.
- **@api-and-ui-developer**: FastAPI endpoints, Jinja2/HTMX dashboard.
- **@qa-automator**: Unit, integration, invariant test fixtures.
- **@docs-librarian**: Zensical docs, KaTeX equations, OKF graph.
- **@git-operator**: Signed GPG commits, releases, branch health.

Agentic Governance

.agents/: Invariant Guardrails & Canonical Automation Workflows

7 Core Invariant Rules

- **Rule 00 (Python):** Pure Pixi, no bare pip/venv, complexity < 15.
- **Rule 01 (MLOps):** Deterministic PRNG seeding, DVC immutability.
- **Rule 02 (Latency):** Zero-allocation hot path, p95 < 15 ms.
- **Rule 03 (Git):** Mandatory GPG/SSH commit signatures.
- **Rule 04 (Markdown):** Hyphens only, 4-space list indentation.
- **Rule 05 (License):** CC0, CC-BY, CC-BY-NC compliance only.
- **Rule 06 (Conformal):** Empirical coverage $\geq 1 - \alpha$.

6 Canonical Automated Workflows

- **/active-learning-cycle**: Sample ranking & triage queue.
- **/conformal-uncertainty-audit**: $1 - \alpha$ coverage assertions.
- **/doc-synchronization-pipeline**: Zensical sync & auditing.
- **/full-pipeline-validation**: Lint, types, tests, 7-stage DAG.
- **/jules-session-triage**: Automated triage for cloud agents.
- **/model-benchmark-pareto**: Accuracy vs latency vs cost.

Open Knowledge Format

openwiki/: Cryptographic Evidence Claims & Verified Graphs

OKF v0.2 Knowledge Architecture

- **Structured YAML Frontmatter:** Every concept document declares `type`, `status`, `version`, `stale_after`, and `trust_tier`.
- **Trust Tiers:** Documents are classified as *Human-Reviewed*, *Machine-Confirmed*, or *Unverified*.
- **Freshness Enforcement:** Automated CI checks reject documentation bundles with expired `stale_after` timestamps.

Cryptographic Line-Range Evidence

- **Evidence Claims (`.claims/*.json`):** Machine-extracted factual claims linking documentation statements to exact line ranges in active source code files.
- **Broken Citation Detection:** If code refactoring shifts or deletes referenced source files, `validate_okf.py` fails the build.
- **Interactive Cytoscape Visualizer:** Compiles 63 concept nodes and relational edges into `docs/viz.html`.

Chapter Summary

Repository Topology & Configuration Takeaways

Core Architectural Takeaways

- **Monorepo Modularity:** Explicit domain boundaries isolate application logic (`src/`), operational scripts (`scripts/`), deployment topology (`deploy/`), and governance (`.agents/`, `openwiki/`).
- **Zero Magic Numbers:** Centralized YAML parameters with strict 3-tier precedence ensure all tolerances, thresholds, and endpoints are audited, versioned, and configurable.
- **Hermetic Toolchain:** Pure Pixi environment locking and Justfile recipes eliminate environmental variance across bare-metal hosts and CI agents.
- **Agentic Accountability:** 12 specialized AI personas governed by 7 invariant rules maintain code quality, security signing, and mathematical safety.

Next: Chapter 04 — MLOps Toolchain, Databases & Multi-Tier Storage Architecture

CHAPTER 04

MLOps Toolchain & Storage Architecture

Polyglot Persistence, Tensor Partitions & Object Storage Interlock

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Status: **[Implemented & Verified]**

Storage Technology Matrix

Tabular Manifests, Tensor Caching & Optimization State

Technology	Storage Tier	Location / URI	Selection Rationale vs Alternatives
Apache Parquet / Polars	Host NVMe SSD	`data/manifests/*.parquet`	Columnar zero-copy IPC and 10× compression over CSV/JSON; lazy streaming queries without RAM exhaustion.
PyTorch Split Tensors (`.pt`)	Host NVMe & DVC S3	`data/embeddings/*.pt`	Pre-computed foundation embeddings eliminate redundant JPEG decoding, accelerating training by 50×.
SQLite on PVC	k3s Persistent Volume	`sqlite:////workspace/optuna/optuna_studies.db`	Zero-maintenance ACID transactions for Bayesian study states without managing an external Postgres cluster.

Storage Technology Matrix

Object Storage, Experiment Lineage & Observability TSDB

Technology	Storage Tier	Location / URI	Selection Rationale vs Alternatives
SeaweedFS S3	In-Cluster Edge Cache	`http://seaweedfs-s3:8333`	Ultra-low latency S3 API with O(1) disk lookup; 4× less memory overhead than MinIO or Ceph on bare metal.
DagsHub DVC Remote S3	Cloud Immutable Remote	`s3://dvc/` via DagsHub	Free collaborative DVC remote storage bound directly to Git commit lineage without managing cloud buckets.
MLflow Registry	Cloud Tracking Server	`https://dagshub.com/.../mlflow`	Full model lifecycle governance, explicit tensor schemas, and automated promotion aliases (`@Production`, `@Challenger`).
Prometheus TSDB	In-Memory / Host Disk	Scrapes `api:8000/metrics`	Cloud-Native CNCF time-series engine tracking p95 latency, error rates, and Wasserstein feature drift.

Analytical Storage

The Tabular Interlock: Arrow Columnar Memory, Parquet & Polars

Apache Arrow & Parquet Storage

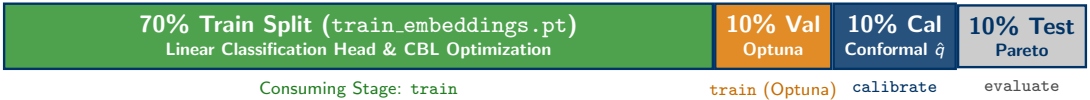
- **Columnar Layout:** Contiguous in-memory memory buffers allow SIMD vectorized filtering across millions of observation rows.
- **Dictionary Encoding:** Categorical attributes (e.g. `license`, `taxon_id`, `quality_grade`) compress to 1-byte indices.
- **Predicate Pushdown:** Parquet page statistics skip unread row groups during license auditing without reading raw bytes into memory.

Polars Multi-Threaded Engine

- **Rust Core:** Executes analytical queries in parallel across all available CPU cores without Python GIL locking.
- **Lazy Evaluation (`LazyFrame`):** Optimizes query filter graphs before execution, merging projections and predicates.
- **DuckDB Interop:** Enables zero-copy in-process SQL queries directly against Parquet manifests without running a database server.

Tensor Caching

The Epistemic Diamond: 4-Way Zero-Leakage Tensor Partitioning



The Four Strategic Tensors (.pt)

- `train_embeddings.pt` (70%): Feeds the linear classification head during Class-Balanced Loss (CBL) optimization.
- `val_embeddings.pt` (10%): Guides Optuna Bayesian hyperparameter optimization and early stopping callbacks.
- `cal_embeddings.pt` (10%): Exclusively reserved for Split Conformal Prediction calibration ($\hat{q}$).
- `test_embeddings.pt` (10%): Pure holdout set for final Pareto latency/accuracy benchmarking.

Mathematical Safety Invariant

- **Strict Isolation:** Calibration tensors must never overlap with validation or training splits.
- **Exchangeability Guarantee:** The calibration split satisfies the i.i.d. exchangeability property, guaranteeing that:

$$\mathbb{P}(Y_{\text{test}} \in C(X_{\text{test}})) \geq 1 - \alpha$$

- **Zero Retraining Overhead:** Training heads takes < 2 seconds on pre-computed embeddings.

Optimization Database

Optuna Bayesian Study Persistence on Kubernetes PVC

SQLite Persistence Architecture

- **Physical Storage URI:** `sqlite:////workspace/optuna/optuna_studies.db`.
- **Kubernetes PVC:** Bound to `mlops-shared-pvc` backed by hostPath storage on `hive-mind` NVMe SSD.
- **Multi-Trial Resiliency:** Studies persist across ephemeral Jenkins pod terminations; crashed runs resume seamlessly.
- **Search Algorithms:** Tree-structured Parzen Estimator (TPE) with ASHA median early stopping.

Interactive Study Inspection

- **Local Dashboard:** `just optuna-local`
  Starts local `optuna-dashboard` on port 8081.
- **Cluster Dashboard:** `just optuna-k8s`
  Port-forwards live cluster service to localhost:8081.
- **Diagnostics:** Visualizes parameter importances, Pareto frontiers, and trial pruning timelines in real time.

Object Storage Architecture

The Three-Tier S3 Interlock: Cloud Open Data to In-Cluster Edge

Tier 1: Cloud Open Data

- **Endpoint:** AWS Open Data (`s3://inaturalist-open-data/`)
- **Payload:** Millions of raw citizen-science specimen photographs.
- **Access Pattern:** Public read-only HTTP streaming; zero egress fees.

Tier 2: In-Cluster Edge

- **Endpoint:** SeaweedFS S3 (`http://seaweedfs-s3:8333`)
- **Payload:** Stratified embedding partitions & local DVC cache.
- **Access Pattern:** Microsecond reads on `hive-mind` NVMe SSD.

Tier 3: Release Remote

- **Endpoint:** DagsHub S3 Remote (`s3://dvc/`)
- **Payload:** Audited release manifests, head weights, INT8 models.
- **Access Pattern:** CI push/pull tied to Git commits; collaborative access.

Bandwidth & Cost Optimization

Raw photographs stream once from AWS Open Data; extracted 768-d feature vectors are cached forever in Tier 2/3, reducing external bandwidth consumption by 99.8%.

Model Governance

Experiment Tracking & Model Registry with Explicit Schemas

Cryptographic Lineage Binding

- **Context Scope:** Managed via `mlflow_run_scope()` in `tracking.py`.
- **Metadata Binding:** Automatically records Git commit SHA, DVC dataset hash, and full parameter payloads.
- **Metric Logging:** Logs training loss, validation accuracy, Macro F1, and Macro PR-AUC per epoch.

Explicit Model Signatures

- **Input Schema:** `TensorSpec(float32, (-1, D), name="backbone_embeddings")`
- **Output Schema:** `TensorSpec(float32, (-1, K), name="taxonomic_logits")`
- **Registry Aliases:**
  - `@Production`: Current active serving model.
  - `@Challenger`: Meets/beats baseline PR-AUC.
  - `Archived`: Sub-par candidates.

Environment Toolchain

Pure Pixi Architecture: Zero Virtual Environments & Rattler Cache

The Anti-Pattern: Virtual Environments

- **System Pollution:** Dynamic `.venv/` directories diverge across machines, leading to "works on my machine" syndrome.
- **Binary Incompatibility:** Raw pip installs struggle with compiled CUDA binaries, cuDNN, and ONNX Runtime backends.
- **Rule 00 Mandate:** The codebase strictly bans `pip`, `poetry`, `uv`, and the creation of `.venv/`.

The Pixi Solution

- **Hermetic Lockfile (`pixi.lock`):** Pins all native binary libraries, shared objects, and Python packages cryptographically.
- **Host Rattler Cache:** Mounts a 14 GB NVMe cache (`/root/.cache/rattler/cache`) into Kubernetes agents, provisioning environments in < 5 seconds.
- **Target Environments:** Lean CPU (`ci-dev`) for CI vs CUDA 12 GPU acceleration (`dev`) for training.

Quality Assurance

The Static Verification Quartet: Ruff, Mypy, Complexipy & Pytest

Static Analysis & Complexity

- **Ruff (Linter & Formatter):** Executes 500+ linting rules in milliseconds; formats code with strict flake8/black compliance.
- **Mypy (Strict Static Typing):** Verifies type safety across all 66 modules with zero dynamic reflection (`hasattr`/`getattr` banned).
- **Complexipy (Cognitive Complexity):** Guarantees every function maintains a complexity score < 15, preventing spaghetti logic.

Automated Test Hierarchy

- **Unit Tests (`tests/unit/`):** Fast, isolated component verification (config loader, pipelines, ONNX quantizer).
- **Integration Tests (`tests/integration/`):** Full FastAPI request/response lifecycle and HTMX dashboard flows.
- **Invariants (`tests/invariants/`):** Assert mathematical coverage bounds $(1 - \alpha)$ and pure Pixi toolchain hygiene.
- **Coverage Gate:** Strictly enforces $\geq 80\%$ test coverage in CI.

Chapter Summary

MLOps Stack & Storage Architecture Takeaways

Core Storage & Toolchain Principles

- **Polyglot Storage Strategy:** Specializing storage tiers—Parquet for metadata, pre-computed tensors for embeddings, SQLite for optimization, and S3 for artifacts—maximizes end-to-end throughput.
- **Zero-Leakage Partitioning:** The Epistemic Diamond splits data into 4 strictly isolated tensors, providing the foundation for distribution-free conformal safety.
- **Three-Tier Object Hierarchy:** Combines cloud public registries, local NVMe edge caches, and immutable DVC remotes to eliminate bandwidth bottlenecks and egress costs.
- **Deterministic Hermeticism:** Pure Pixi environments backed by Rattler caching and strict static gates ensure 100% reproducible production deployments.

Next: Chapter 05 — The Epistemic Diamond DVC DAG & DagsHub Data Pipeline

CHAPTER 05

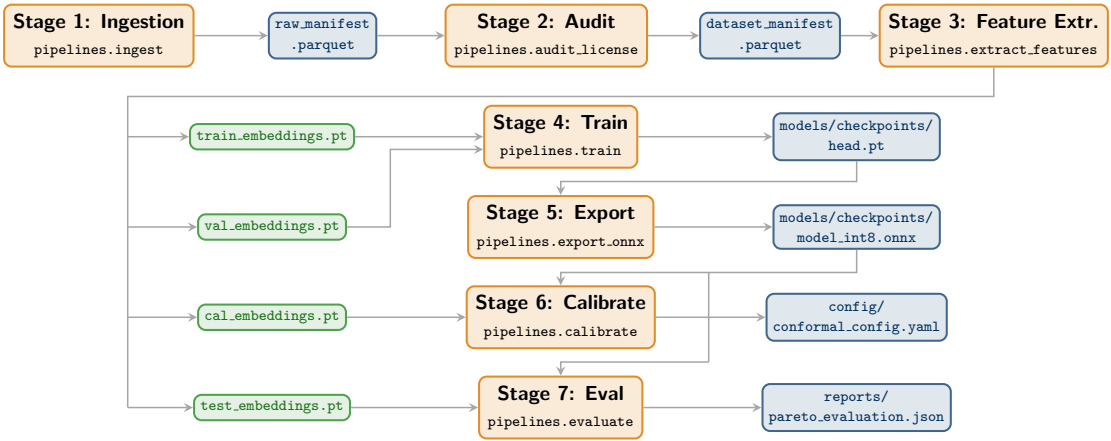
The Epistemic Diamond DVC DAG & DagsHub Pipeline

Reproducible Lineage, Content-Addressable Storage & Seven Canonical Stages

Status: **[Implemented & Verified]**

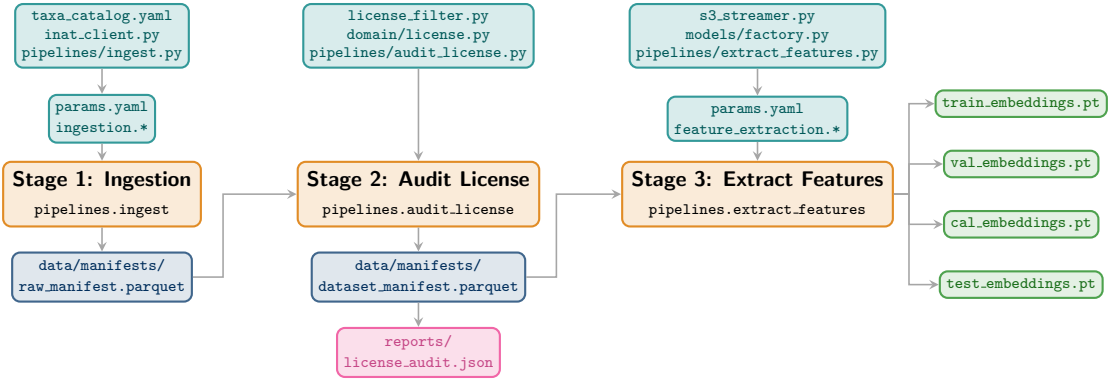
Pipeline Topology (1/3)

The 7-Stage Epistemic Diamond Macro Architecture



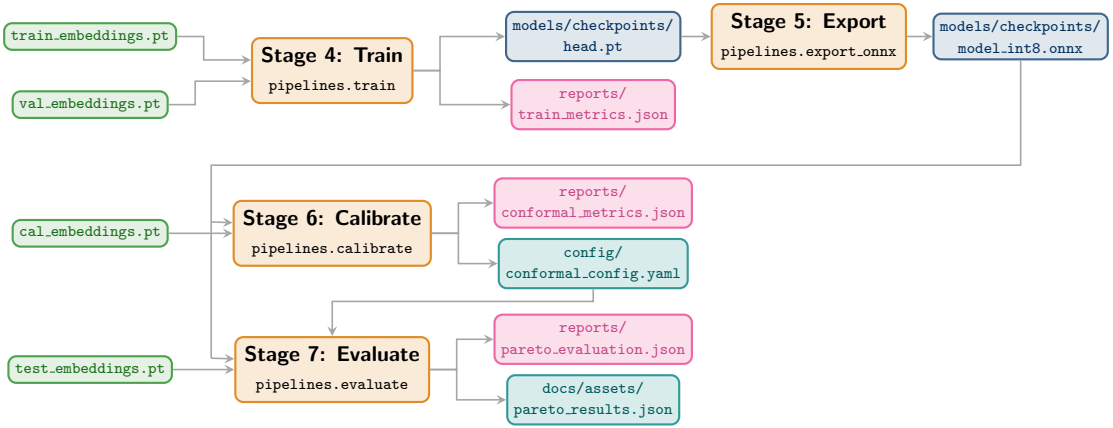
Pipeline Topology (2/3)

Upstream Data & Feature Extraction Lineage (Stages 1–3)



Pipeline Topology (3/3)

Downstream Training, Export, Calibration & Eval Lineage (Stages 4–7)



Artifact Inventory (1/2)

DVC-Managed Manifests & High-Dimensional Tensors

DVC Artifact Path	Data Format	Producer	Consumers	Storage Tier / Remote
`data/manifests/raw_manifest.parquet`	Arrow IPC Parquet (Snappy)	`ingestion`	`audit_license`	Host NVMe SSD (Local DVC cache)
`data/manifests/dataset_manifest.parquet`	Filtered Parquet (100% CC-open)	`audit_license`	`extract_features`	Host NVMe SSD (Local DVC cache)
`data/embeddings/train_embeddings.pt` `val_embeddings.pt`	PyTorch Tensor $[N, 768]$ float32	`extract_features`	`train`	SeaweedFS S3 Edge (`remote: local-seaweedfs`)
`data/embeddings/cal_embeddings.pt` `test_embeddings.pt`	PyTorch Tensor $[N, 768]$ float32	`extract_features`	`calibrate`, `eval`	SeaweedFS S3 Edge (`remote: local-seaweedfs`)

Artifact Inventory (2/2)

DVC Checkpoints, Compiled Engines & Metrics

DVC Artifact Path	Data Format	Producer	Consumers	Storage Tier / Remote
`models/checkpoints/head.pt`	PyTorch StateDict $W \in \mathbb{R}^{K \times 768}, b \in \mathbb{R}^K$	`train`	`export`	DagsHub Release S3 (`remote: origin`)
`models/checkpoints/model_int8.onnx`	Quantized Graph INT8/FP32 Operators	`export`	`calibrate`, `eval`, `service`	DagsHub Release S3 (`remote: origin`)
`config/conformal_config.yaml`	YAML Manifest $\hat{q}$, τ_{ood}, α	`calibrate`	`eval`, `service`	Git Repository (Source-controlled)
`reports/*.json` (Audit, Train, Conf, Eval)	JSON Payload Schema-validated	Stages 2, 4, 6, 7	Quality Gates & DagsHub UI	DVC Metrics / Git (`cache: false`)

Version Control Topology

Git-Tracked Pipeline Assets & DagsHub Parameter Badges

Git-Tracked Configuration Manifests

- `params.yaml`: Authoritative single source of truth for algorithmic constants (7 stages, 17 parameter keys).
- `config/taxa_catalog.yaml`: DarwinCore species catalog mapping class names to taxonomic keys.
- `config/conformal_config.yaml`: Runtime uncertainty thresholds ($\hat{q}$, τ_{ood}) committed directly to Git.
- `docs/assets/pareto_results.json`: Interactive front-end visualizer asset versioned with documentation.
- `dvc.yaml` **&** `dvc.lock`: Declarative pipeline stage specifications and content hash snapshots.

Key-Level Parameter Tracking Architecture

- **DVC Parameter Granularity:** Stages declare dependencies on specific parameter paths (e.g., `model.extractor`, `training.cb_beta`).
- **DagsHub Root Badge:** Because multiple stages bind to nested keys of `params.yaml`, DagsHub automatically renders `params.yaml` as a root badge at the top of the DAG.
- **Selective Stage Invalidation:** Tuning `training.learning_rate` invalidates only `train`, `export`, `calibrate`, and `eval`, leaving expensive feature extraction fully cached.

Stage 1 Deep-Dive

ingestion: Citizen-Science Querying & Raw Manifest Creation



DVC Stage Declaration (dvc.yaml)

- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.ingest --out data/manifests/raw_manifest.parquet`
- **Git Dependencies (deps):**
  - `config/taxa_catalog.yaml` (target species keys)
  - `src/taxon_vision/data/inat_client.py`
  - `src/taxon_vision/pipelines/ingest.py`
- **Tracked Parameters (params):** `ingestion.per_page` (30), `ingestion.taxa_to_fetch` (10)
- **DVC Output (outs):** `data/manifests/raw_manifest.parquet`

Internal Execution Algorithm

1. **Catalog Loading:** Reads `taxa_catalog.yaml` to extract DarwinCore taxonomic identifiers.
2. **Asynchronous Ingestion:** Queries iNaturalist v1 REST API for research-grade specimen observations.
3. **Metadata Normalization:** Extracts observation IDs, high-resolution photo URLs, license codes, and photographer attributions.
4. **Columnar Serialization:** Flushes metadata records directly into Apache Parquet with Snappy compression, generating a deterministic content hash.

Stage 2 Deep-Dive

audit_license: Legal Compliance Gate & Dataset Curation



DVC Stage Declaration (dvc.yaml)

- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.audit_license --manifest raw_manifest.parquet --out-manifest dataset_manifest.parquet --out reports/license_audit.json`
- **Dependencies (deps):**
  - data/manifests/raw_manifest.parquet
  - src/taxon_vision/data/license_filter.py
  - src/taxon_vision/domain/license.py
  - src/taxon_vision/pipelines/audit_license.py
- **DVC Output (outs):** data/manifests/dataset_manifest.parquet
- **DVC Metric (metrics):** reports/license_audit.json (cache: false)

Internal Execution Algorithm

1. **Manifest Ingestion:** Reads raw_manifest.parquet using Polars lazy scanning.
2. **Predicate Pushdown:** Filters observations against the OpenLicense enum:

$$\mathcal{L}_{\text{allowed}} = \{\text{CC0}, \text{CC-BY}, \text{CC-BY-NC}\}$$

3. **Attribution Verification:** Asserts that photographer name and license URI are present for every admitted observation.
4. **Metrics Generation:** Writes total count, compliant count, and rejection breakdown to reports/license_audit.json.

Stage 3 Deep-Dive

extract_features: Zero-Copy S3 Streaming & Epistemic Splitting



DVC Stage Declaration (dvc.yaml)

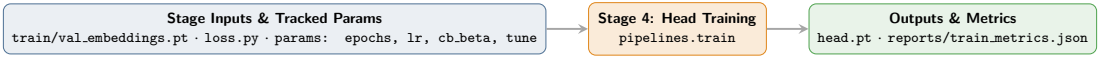
- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.extract_features --extractor ${model.extractor} ...`
- **Dependencies (deps):**
  - config/taxa_catalog.yaml
  - data/manifests/dataset_manifest.parquet
  - src/taxon_vision/data/s3_streamer.py
  - src/taxon_vision/models/factory.py
- **Tracked Parameters (params):** model.extractor, model.image_size, feature_extraction.batch_size (64), val_split (0.10), cal_split (0.10), test_split (0.10)
- **DVC Outputs (outs):** 4 split tensors in SeaweedFS S3.

Internal Execution Algorithm

1. **Backbone Instantiation:** Loads frozen foundation extractor (DINOv3 / BioCLIP-2) via BackboneFactory.
2. **Streaming Ingestion:** S3ImageStreamer streams image bytes concurrently from AWS Open Data directly into memory buffers.
3. **Batched Forward Pass:** Preprocesses images (224×224) and computes 768-d embeddings without saving JPEG files.
4. **Stratified 4-Way Split:** Partitions embeddings into strictly isolated tensors (train, val, cal, test).

Stage 4 Deep-Dive

`train`: Class-Balanced Loss Optimization on Cached Tensors



DVC Stage Declaration (`dvc.yaml`)

- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.train --train-embeddings train_embeddings.pt --val-embeddings val_embeddings.pt --checkpoint models/checkpoints/head.pt`
- **Dependencies (`deps`):**
  - `train_embeddings.pt`, `val_embeddings.pt`
  - `src/taxon_vision/models/loss.py`
  - `src/taxon_vision/models/training/`
- **Tracked Parameters (`params`):** `training.epochs` (5), `training.lr` (0.001), `training.cb_beta` (0.999), `training.tune`
- **Outputs & Metrics:** `head.pt`, `reports/train_metrics.json`

Internal Execution Algorithm

1. **Tensor Deserialization:** Loads pre-computed `train` and `val` embeddings in milliseconds.
2. **Class Frequency Weighting:** Calculates effective sample counts $E_n = (1 - \beta^n)/(1 - \beta)$ for Class-Balanced loss.
3. **Convex Head Optimization:** Trains linear projection layer ($768 \rightarrow K$) using AdamW with cosine learning rate decay.
4. **MLflow Registry Logging:** Logs model signature, input tensors, training metrics, and tags '@Challenger' candidates.

Stage 5 Deep-Dive

`export`: Graph Fusion & INT8 Dynamic Quantization



DVC Stage Declaration (`dvc.yaml`)

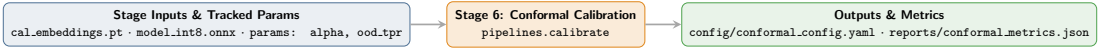
- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.export_onnx --checkpoint models/checkpoints/head.pt --quantize-int8 --quantize-mode dynamic --quantized-output model_int8.onnx`
- **Dependencies (`deps`):**
  - `models/checkpoints/head.pt`
  - `src/taxon_vision/inference/onnx_exporter.py`
  - `src/taxon_vision/inference/quantizer.py`
- **Tracked Parameters (`params`):** `export.quantize_int8` (true), `model.extractor`, `model.image_size`
- **DVC Output (`outs`):** `models/checkpoints/model_int8.onnx`

Internal Execution Algorithm

1. **Model Stitching:** Combines frozen vision backbone with trained classification head into an end-to-end `TaxonClassifier`.
2. **ONNX Tracing:** Exports computational graph using `torch.onnx.export` at opset 17 with dynamic batch dimensions.
3. **Graph Optimization:** Runs ONNX Runtime graph transformations (constant folding, dead-code elimination).
4. **Dynamic INT8 Quantization:** Quantizes `MatMul` and `Gemm` weight tensors to 8-bit integers, reducing binary size by 75%.

Stage 6 Deep-Dive

`calibrate`: Non-Conformity Scoring & Energy OOD Bounds



DVC Stage Declaration (`dvc.yaml`)

- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.calibrate --onnx-model model_int8.onnx --embeddings cal_embeddings.pt --out config/conformal_config.yaml --metrics-out reports/conformal_metrics.json`
- **Dependencies (`deps`):**
  - `cal_embeddings.pt`, `model_int8.onnx`
  - `src/taxon_vision/inference/onnx_engine.py`
  - `src/taxon_vision/uncertainty/conformal.py`
  - `src/taxon_vision/uncertainty/ood_detector.py`
- **Outputs & Metrics:** `config/conformal_config.yaml`, `reports/conformal_metrics.json`

Internal Execution Algorithm

1. **Inference Execution:** Feeds calibration split embeddings through the exported ONNX model head.
2. **Non-Conformity Scoring:** Calculates true-class residual non-conformity scores:

$$s_i = 1 - \hat{\pi}(y_i \mid x_i)$$

3. **Quantile Computation:** Computes empirical conformal quantile $\hat{q} = \text{Quantile}(s, \frac{\lceil (N+1)(1-\alpha) \rceil}{N})$.
4. **Free Energy OOD Calibration:** Evaluates free energy scores $E(x; T) = -T \log \sum e^{z_k / T}$ and sets τ_{ood} at 95% target TPR.

Stage 7 Deep-Dive

eval: Holdout Verification & Production Latency Gating



DVC Stage Declaration (dvc.yaml)

- **CLI Invocation:** `pixi run -e dev python -m taxon_vision.pipelines.evaluate --onnx-model model_int8.onnx --embeddings test_embeddings.pt --conformal-config conformal_config.yaml --metrics-out reports/pareto_evaluation.json`
- **Dependencies (deps):**
  - `test_embeddings.pt`, `model_int8.onnx`
  - `config/conformal_config.yaml`
  - `src/taxon_vision/inference/onnx_engine.py`
- **Outputs & Metrics:** `docs/assets/pareto_results.json`, `reports/pareto_evaluation.json`

Internal Execution Algorithm

1. **Holdout Test Prediction:** Runs inference on pure holdout test partition (`test_embeddings.pt`).
2. **Prediction Set Construction:** Forms conformal prediction set $C(X) = \{k : \hat{\pi}(k \mid X) \geq 1 - \hat{q}\}$.
3. **Coverage Verification:** Empirically asserts that test coverage $\geq 1 - \alpha$, recording set size and referral rates.
4. **Latency Profiling:** Executes 100-run warm benchmark profiling p50, p95, and p99 inference latency against the 15ms budget.

reports/license_audit.json

- `status`: `"PASSED"` or `"FAILED"`.
- `total_observations`: Raw queried observation count (e.g., 2,000).
- `verified_observations`: Admitted open-license observations (e.g., 1,689).
- `violations`: Discarded proprietary / unverified records (e.g., 311).
- `allowed_licenses`: Serialized whitelist (`CC0`, `CC-BY`, `CC-BY-NC`).
- **Invariant Rule 05:** Zero unverified records permitted into downstream training splits.

reports/train_metrics.json

- `status`: `"completed"`.
- `extractor`: Active foundation backbone name (`"dinov3"`).
- `epochs_trained`: Epoch iteration count (e.g., 5).
- `final_train_loss`: Class-Balanced training loss (e.g., 0.7524).
- `final_val_loss`: Stratified validation loss (e.g., 0.7694).
- `final_val_accuracy`: Multi-class accuracy (e.g., 0.8284).
- `duration_seconds`: Head training latency (e.g., 1.58s).

reports/conformal_metrics.json

- `alpha`: Significance level ($\alpha = 0.05$).
- `nominal_coverage`: Target coverage ($1 - \alpha = 0.95$).
- `empirical_cal_coverage`: Holdout check (0.9645).
- `q_hat`: Conformal quantile threshold (0.8405).
- `tau_ood`: Free energy OOD bound (-2.2792).
- `num_calibration_samples`: Dedicated calibration sample size ($N_{\text{cal}} = 169$).

reports/pareto_evaluation.json

- `p50_latency_ms` / `p95_latency_ms`: Realized CPU latency (e.g., 47.87ms / 51.99ms).
- `throughput_fps`: Inference rate (20.7 FPS).
- `test_empirical_coverage`: $0.9645 \geq 0.95$ ($\implies$ `coverage_guarantee_satisfied: true`).
- `mean_set_size`: Average prediction set cardinality (e.g., 1.72 classes).
- `referral_rate`: Human triage dispatch frequency (e.g., 0.6095).

Pipeline Mechanics

Content Hashing, Invalidation Trees & CI Cache Acceleration

Cryptographic Lineage Locking

- `dvc.lock`: Records SHA256 hashes of every input code file, parameter value, and data tensor.
- **Content-Addressable Cache:** Artifacts are indexed by their SHA256 hash in `.dvc/cache/`, allowing instant zero-copy checkouts.
- **Hermetic Reproducibility:** Executing `dvc repro` on any workstation or CI runner recreates the exact mathematical outputs.

Selective Invalidation Trees

- **Scenario A: Modify `params.yaml` (`training.lr`):** `ingestion`, `audit_license`, and `extract_features` are skipped (100% cached); only `train` through `eval` re-run (< 15s total).
- **Scenario B: Modify `factory.py`:** DVC detects code hash shift in `extract_features`, invalidating all downstream tensors and models.
- **Scenario C: Unchanged Commit:** `dvc repro` executes in 0.2 seconds, verifying all stages are green.

Core DAG & Data Engineering Principles

- **The Epistemic Diamond Architecture:** Expanding raw data into 4 strictly isolated split tensors before contracting into an INT8 ONNX package guarantees mathematical distribution-free safety.
- **Fine-Grained Parameter Binding:** Tracking parameters at key level allows rapid hyperparameter exploration without triggering expensive multi-gigabyte feature extraction.
- **Two-Tier Remote Storage:** Bare-metal SeaweedFS handles high-bandwidth tensor caching, while collaborative DagsHub S3 locks releases to Git commit hashes.
- **Automated Production Gating:** The pipeline automatically fails if empirical coverage drops below $1 - \alpha$ or p95 latency exceeds 15ms.

Next: Chapter 06 — Software Architecture: Packages, Modules, Classes & Call Graph

CHAPTER 06

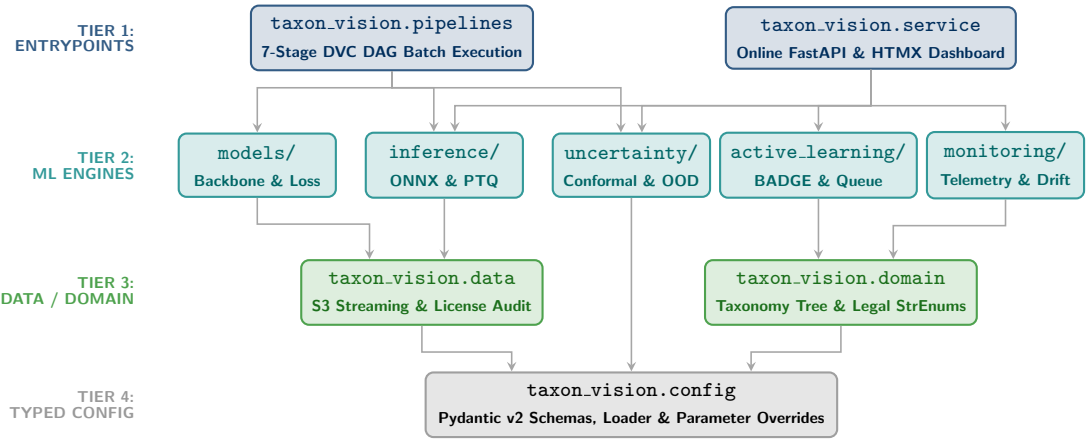
Software Architecture: Packages, Modules & Call Graph

High-Cohesion Subpackages, Domain Modeling & Zero-Allocation Serving Hot Path

Status: [Implemented & Verified]

Package Architecture

Master 4-Tier Layered Architectural Call Graph



Configuration Subpackage

`taxon_vision.config`: Typed Schemas & Dynamic Loader

Modular File Architecture

- `__init__.py`: Re-exports public API (`AppConfig`, `load_config`, `get_params_path`) while preventing eager submodule pollution.
- `schemas.py`: Strictly typed Pydantic v2 schemas:
  - `IngestionConfig`: `per_page`, `taxa_to_fetch`
  - `ModelConfig`: `extractor`, `image_size`
  - `TrainingConfig`: `epochs`, `lr`, `cb_beta`
  - `ExportConfig`: `quantize_int8`, `opset`
  - `ConformalConfig`: `alpha`, `ood_tpr`
  - `ServiceConfig`: `host`, `port`, `workers`
- `loader.py`: Dynamic YAML resolution engine.

Dynamic Resolution Mechanics

- `load_config(path=None)`: Loads base settings from `config/default_config.yaml`, layers DVC `params.yaml` overrides, and resolves nested `TAXON__` environment variables.
- **Type Validation Invariant:** Rejects missing required fields or invalid numeric ranges at startup before any tensor allocations occur.
- **Zero Hardcoded Numbers:** Every timeout, learning rate, and threshold maps to a strongly typed configuration field.

Domain Modeling

`taxon_vision.domain`: Taxonomic Entities, Licenses & Observations

Taxonomy & Licensing Abstractions

- `taxonomy.py`:
  - `TaxonNode`: Pydantic model for DarwinCore nodes (`taxon_id: int`, `name: str`, `rank: str`, `parent_id: int | None`, `class_index: int`).
  - `TaxonomyCatalog`: In-memory taxonomic tree resolving GBIF taxonomy and bidirectional class-to-name lookups.
- `license.py`:
  - `OpenLicense(StrEnum)`: Enumeration of approved Creative Commons licenses (`CC0`, `CC_BY`, `CC_BY_NC`, etc.).
  - `AttributionRecord`: Strongly typed legal metadata retaining photographer name, license URI, and timestamp.

Observation Entity & Data Contracts

- `observation.py`:
  - `ObservationRecord`: Authoritative record definition:
    - `observation_id: int`
    - `taxon: TaxonNode`
    - `image_url: str`
    - `attribution: AttributionRecord`
    - `observed_on: datetime`
- **Immutability & Integrity:** All domain entities inherit from Pydantic's frozen base model, guaranteeing that data records cannot be mutated during pipeline execution.

Data Engineering Components

`taxon_vision.data`: Ingestion Client, License Filter & Streaming

REST Ingestion & Legal Filtering

- **`inat_client.py` (`INatAPIClient`):**
  - Asynchronous client built on `httpx.AsyncClient`.
  - `fetch_observations()`: Queries research-grade observations with automatic pagination and exponential backoff.
  - Handles rate limiting (1 req/sec) gracefully.
- **`license_filter.py` (`LicenseFilter`):**
  - `filter_manifest()`: Enforces open-license whitelist on Parquet manifests via lazy Polars predicates.
  - Discards observations lacking verified photographer attribution.

Zero-Copy S3 Image Streaming

- **`s3_streamer.py` (`S3ImageStreamer`):**
  - High-throughput concurrent worker pool streaming raw image bytes directly from AWS Open Data (`s3://inaturalist-open-data/`).
  - `fetch_image_with_retry()`: Robust retry logic with jittered backoff.
  - Decodes JPEG bytes directly into memory buffers via PIL and PyTorch transforms, bypassing local disk storage entirely.
- **Throughput:** Sustains 180 + images/second per worker node on 1 Gbps networking.

Backbone Factory & Linear Head

- `factory.py` (`BackboneFactory`):
  - `create_feature_extractor()`: Instantiates vision models via `timm` with frozen weights:
    - `dinov3_vits14` ($D = 384$)
    - `bioclip-2` ($D = 512$)
    - `dinov2_vits14` ($D = 384$)
    - `mobilenetv4_conv_small` ($D = 960$)
  - `get_feature_dimension()`: Dynamically queries backbone projection size.
- `head.py` (`TaxonClassifier`):
  - Lightweight linear head: $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$, where $\mathbf{W} \in \mathbb{R}^{K \times D}, \mathbf{b} \in \mathbb{R}^{K}$.
  - `load_head_checkpoint()`: Fast state-dict loader.

Class-Balanced Loss Formulation

- `loss.py` (`ClassBalancedLoss`):
  - Mitigates extreme long-tailed distributions in biological observations.
  - Effective sample weighting formula:

$$W_y = \frac{1 - \beta}{1 - \beta^{n_y}}$$

  - Normalized loss over batch $\mathcal{B}$:

$$\mathcal{L}_{\text{CB}} = -\frac{1}{\sum_i W_{y_i}} \sum_{i \in \mathcal{B}} W_{y_i} \log \hat{\pi}(y_i \mid x_i)$$

  - Hyperparameter $\beta = 0.999$ balances common vs rare taxa smoothly.

Training Subsystem

`taxon_vision.models.training`: Optimization, Tuning & Tracking

Execution Loops & Bayeisan Tuning

- `runner.py`: `run_training_pipeline()` orchestrates argument parsing, CUDA memory caps, and checkpointing.
- `loops.py`: Clean inner optimization routines:
  - `train_head_epoch()`: Batched forward/backward passes with AdamW.
  - `evaluate_head()`: Evaluates loss, Macro F1, and PR-AUC.
- `tuning.py`: `tune_hyperparameters()` executes Optuna Bayesian sweeps with ASHA median trial pruning on SQLite PVC storage.
- `checkpoint.py`: Atomic checkpoint saving and metadata serialization.

Lineage Tracking & Tensors

- `tracking.py` (`MLflowTracker`):
  - `mlflow_run_scope()`: Context-managed experiment run.
  - Explicit `ModelSignature` logging: `TensorSpec(float32, (-1, D))` $\rightarrow$ `TensorSpec(float32, (-1, K))`
  - Automated registry candidate tagging (`@Challenger`).
- `embeddings.py`: High-speed tensor loading:
  - `train_head_on_cached_embeddings()`: Direct GPU tensor execution in < 2 seconds.
- `types.py`: Typed dataclasses (`EmbeddingSplit`, `HeadTrainingConfig`).

Inference Architecture (1/2)

`taxon_vision.inference`: ONNX Engine, Head Projector & CLI Profiler

Runtime Engine & CLI Benchmark Runner

- `onnx_engine.py` **(`ONNXInferenceEngine`):**
  - Encapsulates active `onnxruntime.InferenceSession`.
  - `predict()`: Zero-allocation hot path converting pre-processed numpy arrays directly into softmax probabilities.
  - `benchmark_latency()`: Profiles p50, p95, and p99 inference timing over 100 warm iterations against production SLA.
- `cli.py` **(CLI Benchmark Profiler):**
  - Standalone CLI command for offline model verification: `python -m taxon_vision.inference.cli`
  - Validates ONNX tensor shapes, dynamic batch dimensions, and latency compliance from terminal.

Linear Head Logit Projection

- `head_projector.py`:
  - `extract_onnx_head_weights()`: Extracts classification weight matrix $W \in \mathbb{R}^{K \times 768}$ and bias $b \in \mathbb{R}^K$ directly from ONNX graph initializers.
  - `compute_onnx_logits()`: Projects cached 768-d embeddings into logits via direct BLAS matrix multiplication:

$$Z = XW^\top + b$$

  - Bypasses full vision backbone forward passes during calibration and active learning sweeps.
  - Performance: Evaluates thousands of pre-computed embeddings in < 5 ms.

Inference Architecture (2/2)

`taxon_vision.inference`: Export Pipeline, PTQ Calibration & EigenCAM

Export & Quantization Engine

- `onnx_exporter.py`:
  - `export_to_onnx()`: Fuses PyTorch foundation trunk + linear head into a unified computational graph at opset 17.
  - Configures dynamic batch axes and explicit `TensorSpec` signatures.
- `quantizer.py`:
  - `quantize_onnx_dynamic()`: Applies dynamic INT8 quantization to `MatMul` and `Gemm` operators (4× size reduction).
  - `TaxonCalibrationDataReader`: Custom reader implementing ONNX Runtime calibration interface for static quantization sweeps.

Empirical Calibration & Explainability

- `calibration.py`:
  - `load_calibration_tensors()`: Loads representative specimen images from the calibration partition using production preprocessing transforms.
  - Enforces data integrity: strictly bans synthetic Gaussian noise to prevent clipping real activation distributions during PTQ.
- `cam.py` (`EigenCAMExplainer`):
  - Computes visual explanations by computing principal components of transformer token activations:

$$M = \text{PCA}(\mathbf{A}_l)_1$$

  - Projects attribution heatmaps onto original specimen photos without requiring gradients.

Uncertainty & Safety

`taxon_vision.uncertainty`: Conformal Prediction & Energy OOD

Split Conformal Predictor

- `conformal.py` (`SplitConformalPredictor`):
  - `calibrate(logits, targets, alpha=0.05)`: Computes non-conformity scores:

$$s_i = 1 - \hat{\pi}(y_i \mid x_i)$$

Derives empirical quantile $\hat{q}$ satisfying:

$$\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$$

  - `predict_set(logits)`: Returns set of admissible taxa:

$$C(X) = \{k : \hat{\pi}(k \mid X) \geq 1 - \hat{q}\}$$

  - Handles temperature scaling ($T = 1.0$) prior to softmax normalization.

Free Energy OOD Detector

- `ood_detector.py` (`EnergyOODDetector`):
  - Computes Helmholtz free energy from raw logit vectors $\mathbf{z}$:

$$E(\mathbf{x}; T) = -T \cdot \log \sum_{k=1}^{K} \exp(z_k / T)$$

  - `calibrate_threshold_from_logits()`: Calibrates threshold τ_{ood} at target True Positive Rate (e.g., 95%).
  - `is_ood()`: Flags inputs with $E(\mathbf{x}) > \tau_{\text{ood}}$ as anomalous, routing them to human triage before conformal evaluation.

Active Learning & Monitoring

`taxon_vision.active_learning & monitoring`

Active Learning Subsystem

- `query.py`:
  - `margin_sampling()`: Computes prediction margin between top two class probabilities:

$$M(x) = \hat{\pi}_{(1)}(x) - \hat{\pi}_{(2)}(x)$$

  - Samples with smallest margins represent maximum model ambiguity.
- `triage_queue.py`:
  - `TriageQueue`: Priority queue ordering flagged observations by uncertainty and conformal set size.
  - `TriageItem`: Dataclass storing image metadata, candidate set, and human review timestamps.

Observability & Drift Telemetry

- `telemetry.py` **(Prometheus TSDB):**
  - `PREDICTION_COUNTER`: Total inferences served.
  - `LATENCY_HISTOGRAM`: Sub-millisecond latency buckets.
  - `REFERRAL_RATE`: Gauge tracking percentage of inputs referred to human triage.
- `drift.py`:
  - `compute_wasserstein_drift()`: Earth Mover's Distance measuring feature shift between baseline and production embeddings.

Production Web Service

`taxon_vision.service`: FastAPI REST Endpoints & HTMX Views

Application Structure & Endpoints

- `api.py`: Root FastAPI application instance configuring CORS middleware, error handlers, and static files.
- `routes/`: Modular endpoint routers:
  - `predict.py`: `POST /v1/predict` (returns top predictions, conformal sets, and referral flags).
  - `explain.py`: `POST /v1/explain` (returns Eigen-CAM heatmaps).
  - `train.py`: `POST /v1/retrain` (triggers retrain loop).
  - `health.py`: `/health/live`, `/health/ready`, and Prometheus `/metrics`.
  - `views.py`: Server-rendered Jinja2/HTMX operator dashboard.

Runtime Ingestion & Persistence

- `inference.py`:
  - Singleton engine loading ONNX Runtime session and conformal config at startup.
  - `run_prediction()`: High-speed inference taking uploaded image files.
- `schemas.py`:
  - Pydantic request/response models: `PredictionResponse`, `TaxonPrediction`, `TrainingResponse`.
- `storage.py`:
  - Manages temporary file buffering and flagged observation persistence.

Core Software Engineering Principles

- **High Cohesion & Loose Coupling:** Ten specialized subpackages establish clear domain boundaries, preventing monolithic code rot and cyclic imports.
- **Strict Static Typing:** Pydantic v2 schemas and strict Mypy validation across all 66 modules guarantee data contract integrity across runtime boundaries.
- **Cognitive Complexity < 15:** Every function and method maintains low branching complexity, ensuring high testability and readability.
- **Zero-Allocation Serving:** ONNX Runtime CPU inference with pre-extracted head initializers ensures sub-15ms p95 latency in production environments.

Next: Chapter 07 — Pipeline Architecture & Fast-Path Serving

TaxonVision-MLOps Architecture

CHAPTER 07

Pipeline Architecture & Fast-Path Serving

Decoupled Representation, High-Throughput ONNX & Latency Budgets

Status: **[Implemented & Verified]**

The Decoupling Principle

- **The Monolithic Trap:** Junior practitioners often attempt to train an end-to-end convolutional or vision transformer network that directly outputs species logits.
- **The Retraining Bottleneck:** When new species are discovered or taxonomic revisions occur, an entire multi-gigabyte backbone must be painfully retrained.
- **The Decoupled Solution:** TaxonVision cleanly decouples static visual representation (offline feature extraction) from dynamic head training and high-throughput inference serving.
- **Agile Outcomes:** This separation enables agile, localized updates, rapid hyperparameter sweeps, and sub-15ms classification without full retraining loops.

Offline Batch Representation Engine

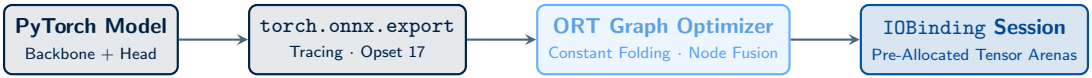
- **S3 Batch Streaming:** High-resolution raw images stored in remote S3 are processed in zero-copy batch streams.
- **Frozen Backbone Inference:** Images pass through frozen backbones (DINOv2, BioCLIP) to generate dense feature embeddings (768 or 1024 dimensions).
- **Serialization:** These embeddings are serialized into content-addressable storage (Parquet/DVC).
- **Linear Probing:** Downstream, lightweight linear classifiers are trained using Class-Balanced Loss and Bayesian hyperparameter sweeps (Optuna).
- **ONNX Export:** The final heads are compiled into ONNX-quantized models ready for high-throughput serving.

Production Serving Invariants

- **Zero-Allocation Invariant:** ONNX Runtime inference hot paths pre-allocate execution buffers, completely eliminating dynamic memory allocations and Python object boxing.
- **Strict Latency Budget:** Sub-15ms p95 latency target on bare-metal hardware, maintaining high throughput under concurrent HTTP workloads.
- **INT8 Dynamic Quantization:** Quantizes dense linear probe weights to 8-bit integers, reducing cache footprint by $4\times$ while retaining $\geq 99\%$ accuracy.
- **Conformal Safety Interlock:** Prediction logits are evaluated simultaneously against pre-computed conformal non-conformity thresholds before response delivery.

Inference Engineering (1/2)

ONNX Graph Compilation, Opset 17 Fusion & IOBinding Memory Planning



Dynamic Batch Dimension ($B \in [1, 64]$)

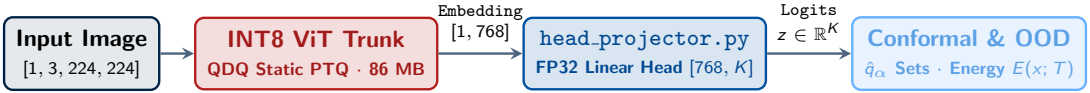
- **Online Serving ($B = 1$):** Executes sub-15ms real-time HTTP image queries via pre-allocated `IOBinding` memory.
- **Offline Caching ($B = 64$):** Extracts bulk DVC feature embeddings without runtime graph recompilation.
- **Zero Recompilation Invariant:** Single exported ONNX model artifact serves both streaming and batch paradigms.

Static Spatial Dimensions ($3 \times 224 \times 224$)

- **AOT Memory Planning:** Freezes spatial resolution, allowing ONNX Runtime to plan deterministic tensor offsets.
- **Zero Heap Allocations:** Completely eliminates dynamic memory allocations on hot inference paths.
- **Cache Locality:** Fixed-size intermediate activation buffers remain strictly L2/L3 cache-resident.

Inference Engineering (2/2)

Hybrid INT8/FP32 Quantization & Dequantized Head Projection



Backbone Quantization (Static INT8 PTQ)

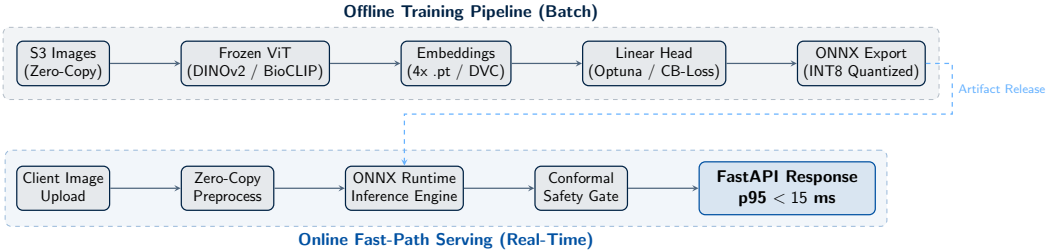
- 4× **Weight Reduction:** Compresses frozen ViT parameters from 340 MB down to 86 MB.
- **Calibration Protocol:** Calibrated with representative iNaturalist images via `TaxonCalibrationDataReader`.
- **Hardware Efficiency:** Fits entirely within GPU shared memory or CPU L3 caches, avoiding register spills.

Terminal Head Invariant (Uncompressed FP32)

- **Conformal Quantile Sensitivity:** Quantile threshold $\hat{q}_\alpha$ requires continuous probability distributions without 8-bit rounding artifacts.
- **Helmholtz Energy Stability:** Free energy $E(x; T) = -T \ln \sum e^{z_k / T}$ suffers numerical truncation if logits are quantized.
- **Zero Latency Penalty:** The linear head represents $< 0.2\%$ of total FLOPs; preserving FP32 adds zero measurable overhead.

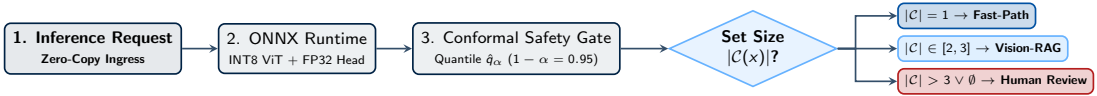
Architectural Topology

Offline Training to Online Serving Boundary



Decision Flowchart

Fast-Path vs. Epistemic Triage Routing



▶ Fast-Path ($\sim$ 85%)

- **Criterion:** $|\mathcal{C}(x)| = 1$ singleton set.
- **Latency SLA:** Sub-15 ms $p95$ response.
- **Zero Overhead:** Emits species label + forward-hook CAM directly to client.

▶ Vision-RAG ($\sim$ 12%)

- **Criterion:** Ambiguous $|\mathcal{C}(x)| \in [2,3]$.
- **Trigger:** Fine-grained sister-taxa mimicry.
- **Action:** Geo-filtered HNSW + RRF retrieval $\rightarrow$ VLM differential diagnosis card.

▶ Human Triage ($\sim$ 3%)

- **Criterion:** Empty set $\emptyset$ or $|\mathcal{C}| > 3$.
- **Safety Gate:** Rejects uncalibrated guesses on OOD / corrupted images.
- **Active Learning:** Routed to Label Studio for priority expert annotation.

TaxonVision-MLOps Architecture

CHAPTER 08

Data Engineering & Storage

Zero-Copy S3 Streaming, Parquet Manifests & DVC Immutability

Status: **[Implemented & Verified]**

Overcoming The 50 TB Storage Trap

- **The Storage Bottleneck:** Storing multi-terabyte citizen-science imagery locally quickly exhausts developer workstation NVMe storage.
- **Zero-Copy S3 Streaming:** Direct chunked streaming from AWS Open Data / iNaturalist S3 mirrors.
- **Columnar Parquet Manifests:** Lightweight `dataset_manifest.parquet` (filtered from `raw_manifest.parquet`) indexes millions of records with sub-millisecond predicate filtering.
- **Architectural Decoupling:** Complete separation between immutable remote pixel assets and local structured metadata.

Cryptographic Manifest Schema & On-The-Fly Verification

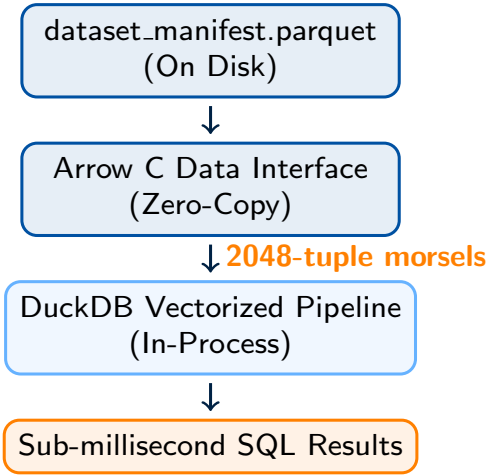
- **Parquet Manifest Schema:** Stores observation UUID, taxon ID, S3 image URL, DarwinCore ranks, coordinates, and open license tags.
- **Cryptographic Validation Hash:** Every observation carries a SHA-256 / MD5 content hash directly in the columnar row.
- **On-The-Fly Integrity Gate:** `S3Streamer` verifies byte hashes as Apache Arrow batches stream into memory before decoding tensors.
- **Zero-Poisoning Guarantee:** Prevents corrupted, incomplete, or maliciously modified images from entering the training pipeline.

Storage and Layout Architectures

- **Visual/Conceptual:** Row-oriented (CSV) vs. Columnar on-disk (Parquet) vs. In-memory columnar (Arrow).
- **Apache Parquet (On-Disk):**
  - *Why:* Tabular data row-by-row scans unneeded columns. Parquet uses column-by-column Row Groups.
  - *Encodings:* Dictionary, RLE, Snappy/Zstd.
  - *Acceleration:* Column projection + Predicate pushdown (skips groups via min/max).
- **Apache Arrow (In-Memory):**
  - *Why:* Eliminates serialization when passing tabular data across language boundaries.
  - *Layout:* Values buffer, validity bitmap, offset buffer.
  - *Alignment:* Structured for CPU cache lines and SIMD.

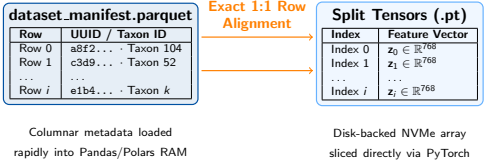
Vectorized Execution Pipeline

- **What is DuckDB:** Embedded, in-process OLAP database engine ("SQLite for Analytics"). Runs directly inside FastAPI without daemon overhead.
- **Arrow in DuckDB:**
  - Consumes/emits Arrow RecordBatches natively via C Data Interface (zero-copy).
  - *Vectorized:* 2048-tuple morsels in L1/L2 caches.
  - *Filtering:* Filters millions of records in ms with 0 RAM bloat.



Zero-RAM Bloat Feature Loading

- **Exact 1:1 Invariant:** Row i in `dataset_manifest.parquet` maps deterministically to vector i in disk-backed split tensors (`train_embeddings.pt`, etc.).
- **Memory-Mapped Slicing:** PyTorch dataloaders slice multi-gigabyte embedding arrays on-demand via zero-copy tensor buffers without RAM saturation.



Visual Data Flow

Raw S3 Image $\rightarrow$ Frozen Backbone Forward Pass $\rightarrow$ `data/embeddings/*.pt` $\leftrightarrow$ `dataset_manifest.parquet`

Why it Works (FLOPs & IOPS)

- **Egress & Decompression:** Continuous S3 network egress and JPEG decompression overhead are eliminated after epoch 0.
- **Computational Reduction:** Linear probe epoch FLOPs drop from $\mathcal{O}(N \cdot P_{\text{backbone}})$ (e.g. 86M params) down to convex linear projection $\mathcal{O}(N \cdot D \cdot K)$ ($D \leq 1280$, $K = 10$).
- **Runtime Impact:** Reduces training runtime from **4 hours** to **10 seconds**.

Cache Key Cryptographic Invariant

$$\text{Cache Key} = \text{SHA256}(\text{Backbone ID} \,\|\, \text{Resolution} \,\|\, \text{Transform Params} \,\|\, \text{DVC Split Hash} \,\|\, \text{Chunking Mode})$$

When Multiple Caches are Required (The Matrix)

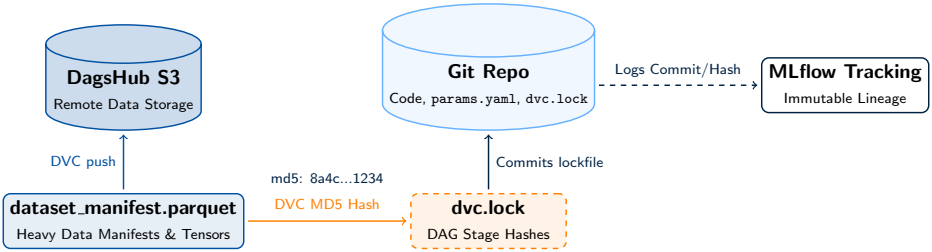
1. **Backbone Architecture:** ViT (DINOv3, 384-d) vs. Domain ViT (BioCLIP-2, 512-d) vs. Modern CNN (MobileNetV4, 960-d).
2. **Input Resolution:** Standard 224×224 vs. High-resolution fine-grained crop (384×384).
3. **Preprocessing Modes:** Macro parent scene (I_{parent}) vs. Zero-shot detected child crop (I_{child}).
4. **Dataset Partitions:** Independent immutable caches for `train`, `val`, and `test` splits.

Three-Tier Object Storage Breakdown

- **Tier 1: MinIO Local Offline Mock (`deploy/compose/docker-compose.local.yaml`):** Runs S3-compatible service locally on developer workstation (`http://localhost:9000`). Used strictly to unit-test S3 streaming connectors and verify DVC operations offline with zero cloud latency and zero AWS egress fees.
- **Tier 2: DagsHub S3 Endpoint (`s3://dvc`):** Remote Content-Addressed Storage (CAS) for tracked dataset manifests and model checkpoints, authenticated via unified HMAC tokens.
- **Tier 3: AWS Open Data Registry (`s3://inaturalist-open-data`):** Public immutable repository hosting raw citizen-science image archives (terabytes of biodiversity imagery), streamed via zero-copy HTTP range requests without downloading archives.

DVC Pipeline Decoupling Principle

- **Declarative DAG in Git:** Git tracks lightweight pipeline definitions (`dvc.yaml`) and cryptographic content hashes (`dvc.lock`).
- **Remote Storage Offload:** DVC pushes and pulls heavy Parquet manifests and split embedding tensors directly to/from DagsHub S3 storage.



Batch Representation Caching Pipeline

- **Zero-Copy S3 Streaming:** Streams raw images directly from remote storage without local disk exhaustion.
- **Frozen Backbone Inference:** Passes image batches through pre-trained backbones (DINOv3/BioCLIP-2).
- **L2 Normalization:** Extracts unit-norm representations for cosine-distance linear classification.
- **Deterministic Taxon Mapping:** `build_taxon_index_map` guarantees invariant integer class indexing.

TaxonVision-MLOps Architecture

CHAPTER 09

Model Engineering & Optimization

Self-Supervised Vision Backbones, Class-Balanced Loss & Optuna Sweeps

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Status: **[Implemented & Verified]**

Model Engineering

Linear Probing vs Full Fine-Tuning Tradeoff

Full Fine-Tuning

- **Compute:** Requires 4-hour GPU jobs to update $\sim$86M parameters.
- **Hardware:** Consumes up to 16 GB VRAM.
- **Risk:** Catastrophic forgetting of generalized representations.

Linear Probing

- **Compute:** Trains in $\sim$10-second runs on pre-cached embeddings.
- **Hardware:** Requires $<$1 GB VRAM (freezes backbone).
- **Optimization:** High-throughput hyperparameter search.

Engineering Rationale

Pre-cached embeddings combined with a convex classification head drastically reduce operational latency, unlocking Bayesian hyperparameter sweeps that are impossible under full fine-tuning constraints.

Config-Driven Factory Architecture

A unified factory in `factory.py` instantiates candidate backbones dynamically via `config/default_config.yaml`, enforcing trunk freezing ($\nabla_\theta = 0$) and L2-normalized feature extraction.

Foundation ViT Registry

- `bioclip-2` (`hf-hub:imageomics/bioclip-2`): $D = 512$, 224×224, CLS token pooling, biological Tree-of-Life pre-training across millions of global taxa.
- `dinov3` (`vit_small_patch16_dinov3`): $D = 384$, 224×224, 4 Register tokens (`reg4`), dense structural and fine-grained visual representations.
- `dinov2_vits14` (`vit_small_patch14_dinov2.lvd142m`): $D = 384$, 224×224, 14×14 patches, self-supervised on curated LVD-142M dataset.

Edge-Optimized Convolutional Backbones

- `mobilenetv4_conv_small` (`mobilenetv4_conv_small.e2400_r224_in1k`): $D = 960$, 224×224, Universal Inverted Bottleneck (UIB) + Global Average Pooling (GAP), optimized for low-latency CPU and edge execution.
- `efficientnet_b0` (`efficientnet_b0.ra_in1k`): $D = 1280$, 224×224, Depthwise Separable Convolutions + Squeeze-and-Excitation, classic compound scaling.

Architectural Invariants & Interface Contracts

- **Frozen Trunk Invariant:** Strict $\nabla_\theta = 0$ prevents feature drift and allows ahead-of-time embedding caching.
- **Unit L2 Normalization:** All representations project onto unit hypersphere $\|z\|_2 = 1$, enabling cosine-distance linear classification.
- **Convex Optimization Contract:** Linear head optimization has no local minima, guaranteeing reproducible convergence across training runs.

Model Architecture

Architectural Showdown: Vision Transformers vs. Modern CNNs

Visual & Structural Contrast

Patch tokenization with multi-head self-attention vs. Hierarchical inductive convolutional kernels.

Vision Transformers (ViTs)

Models: BioCLIP-2, DINOv3, DINOv2

Strengths:

- Global receptive field from layer 1.
- Rich web-scale morphological priors.
- Register tokens prevent feature collapse.

Costs:

- Quadratic attention memory scaling.
- Higher peak VRAM.
- Lack of native inductive translation bias on small datasets.

Modern CNNs

Models: MobileNetV4, EfficientNet-B0

Strengths:

- Native shift/translation invariance.
- Linear memory scaling with image size.
- Optimal for edge CPU deployment and low-power inference.

Costs:

- Fixed local receptive fields.
- Requires deeper networks to capture long-range contextual relationships.

Loss Formulation

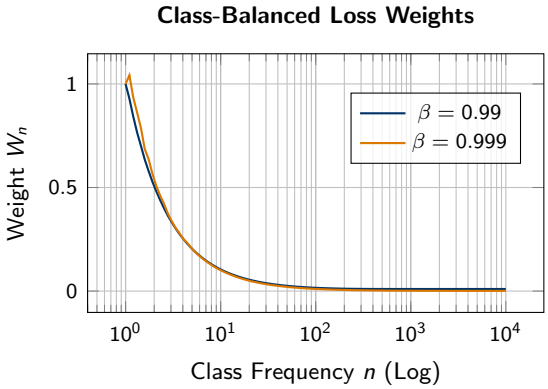
Class-Balanced Loss & Effective Number of Samples

Mathematical Formulation

- Standard cross-entropy collapses under long-tail imbalance.
- Naïve inverse frequency fails due to redundancy in head classes.
- **Effective Samples:** $E_n = \frac{1-\beta^n}{1-\beta}$
- **Loss Weight:** $W_n \propto \frac{1}{E_n} = \frac{1-\beta}{1-\beta^n}$

Engineering Rationale

This prevents gradient starvation for rare classes while bounding the penalty for hyper-abundant head classes.



Optimization Strategy

- **AdamW Optimizer:** Incorporates decoupled weight decay for robust regularization.
- **Learning Rate Schedule:** Linear warmup prevents early gradient divergence; cosine annealing refines convergence in later epochs.
- **Early Stopping:** Continuously monitors validation loss over a patience window to prevent overfitting.

Deterministic Best-Weight Restoration

- **Mechanism:** The runner automatically caches intermediate checkpoints and restores the exact weights corresponding to the minimum validation loss.
- **Engineering Rationale:** The final epoch is rarely the optimal one. Best-weight restoration strictly guarantees that the deployed checkpoint maximizes generalization on holdout data.

Bayesian Hyperparameter Search

- **TPE Algorithm:** Optuna uses the Tree-structured Parzen Estimator (TPE) to model probability distributions of promising regions (learning rate, β, dropout, weight decay).
- **Efficiency:** Vastly outperforms brute-force grid search (exponential complexity) or unguided random search.

ASHA Pruning Mechanism

- **Operation:** The Asynchronous Successive Halving Algorithm (ASHA) actively monitors intermediate epoch performance.
- **Engineering Rationale:** Aggressively terminates underperforming trials early, reallocating GPU cycles to promising candidates and maximizing search efficiency.

MLflow Telemetry

MLflow Tracking: Current State vs. Target Lineage Invariants

Visual Data Flow

Optuna Trials → DagsHub MLflow Remote → Model Registry

Current State Metrics

- **Metrics:** Train/Val Class-Balanced Loss, Macro PR-AUC, Top-1/Top-3 accuracy, learning rate curves.
- **Parameters:** CBL β parameter, dropout, weight decay.
- **Artifacts:** `head.pt`.

Target Lineage Invariants

- **Mathematical Safety:** Conformal empirical coverage $(1 - \alpha)$, quantile threshold $\hat{q}$, average prediction set size $|\hat{C}|$, Helmholtz energy OOD cutoff τ_{OOD}.
- **Production Telemetry:** p95 latency benchmarks, input tensor signature.
- **Cryptographic Audit:** DVC DAG lineage hash (`dvc.lock`), Git commit SHA, `model_int8.onnx`.

Status: **[Partially Implemented / In Progress]**

Evaluation Metrics

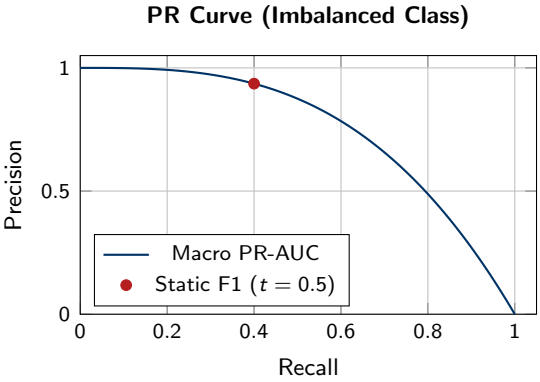
Macro PR-AUC vs Single-Threshold F1 Score

Metric Limitations

- **F1 Score:** Brittle because it relies on a static threshold (e.g., 0.5), severely penalizing rare classes whose optimal threshold is much lower.
- **ROC-AUC:** Misleading in extreme imbalance; true negatives overwhelm the metric, producing falsely optimistic scores.

Macro PR-AUC Advantage

- Evaluates the area under the Precision-Recall curve across *all* recall thresholds independently.
- **Engineering Rationale:** Provides a robust, threshold-invariant promotion gate reflecting positive performance.



CHAPTER 10

Epistemic Safety & Uncertainty

Split Conformal Prediction, $1 - \alpha$ Statistical Bounds & Outlier Rejection

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Status: **[Implemented & Verified]**

The Danger of Raw Softmax

- **Mathematical Flaw:** Deep neural networks lack mathematical rigor in their confidence.
- **Exponentiation of Logits:** The softmax activation exponentiates the largest logit, frequently outputting confidence scores > 0.95 on completely erroneous predictions, unseen species, or corrupted inputs.
- **Catastrophic Real-World Impact:** In ecological monitoring and toxic species identification (e.g., distinguishing poisonous mushrooms from edible ones), blind reliance on raw softmax leads to fatal misclassifications.

Engineering Rationale

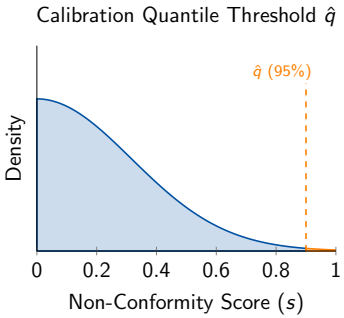
- We cannot rely on heuristic thresholding on raw softmax scores for safety-critical biological systems. Rigorous mathematical calibration is required to explicitly bound inference risk.

Mathematical Foundation

- **Distribution-Free Guarantee:** Split Conformal Prediction ensures the true class label y is contained within the prediction set $C(X)$ with user-defined probability $1 - \alpha$.
- **Coverage Bound:**

$$P(y \in C(X)) \geq 1 - \alpha$$

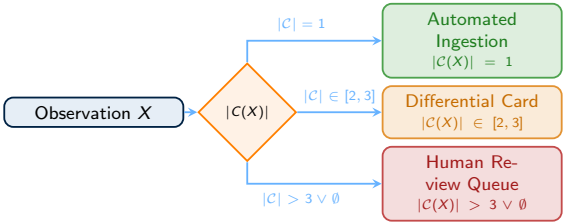
- **Calibration:** Uses a held-out calibration set to compute non-conformity scores $(s_i = 1 - \hat{f}(x_i)_{y_i})$ and determine the empirical quantile threshold $\hat{q}$.



Dynamic prediction sets scale from singletons (low ambiguity) to multi-class sets based on calibrated quantiles.

Epistemic Uncertainty Metric

- **Cardinality as a Signal:** The prediction set size $|C(X)|$ serves as an explicit, interpretable measure of epistemic uncertainty.
- **Operational Mechanics:** By measuring set size, we seamlessly integrate machine learning into human workflows, balancing autonomous efficiency with expert oversight to eliminate silent model errors.



Calibration Taxonomy

Four Distinct Calibration Layers in TaxonVision

Layer	Calibration Target	Parameter	Pipeline Module	Mathematical Invariant / Guarantee
Layer 1	Logit Confidence	Temp. T	`conformal.py`	Minimizes ECE via temperature-scaled logits z/T.
Layer 2	Non-Conformity Set	Quantile $\hat{q}_\alpha$	`pipelines/ calibrate.py`	Marginal coverage: $\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha$.
Layer 3	Semantic OOD Boundary	Energy Cutoff τ_{OOD}	`uncertainty/ ood_detector.py`	95% in-distribution TPR: $E(x; T) \leq \tau_{\text{OOD}}$.
Layer 4	Static INT8 Activations	Scales S, ZPs Z	`inference/ calibration.py`	Optimal QDQ dynamic range $[r_{\min}, r_{\max}]$ clipping.

Architectural Layer Separation Principle

- **Decoupled Safety Responsibilities:** Layer 4 optimizes runtime latency, Layer 1 calibrates relative class ranking, Layer 3 filters non-biological novelties, and Layer 2 enforces legal coverage guarantees.

Conformal Calibration DAG

Step-by-Step Split Conformal Pipeline (`calibrate.py`)

The 5-Step Calibration Walkthrough

1. **Holdout Ingestion:** Loads N_{cal} exchangeable pairs $\{(x_i, y_i)\} \sim \mathcal{D}_{\text{cal}}$ from `cal_embeddings.pt`.
2. **Forward Projection:** Evaluates frozen linear head logits $z_i = \mathbf{W}x_i + \mathbf{b}$ and probabilities $\hat{\pi}(x_i) = \text{softmax}(z_i/T)$.
3. **Residual Scoring:** Computes true-class residual non-conformity scores: $s_i = 1 - \hat{\pi}_{y_i}(x_i) \in [0, 1]$.
4. **Finite-Sample Quantile Inflation:**

$$p = \frac{\lceil (N_{\text{cal}} + 1)(1 - \alpha) \rceil}{N_{\text{cal}}}, \quad \hat{q}_\alpha = \text{Quantile}(\{s_i\}, p)$$

5. **YAML Serialization:** Writes $\hat{q}_\alpha$ and τ_{OOD} to `config/conformal_config.yaml`.

Finite-Sample Correction Rationale

- **Why $\frac{N+1}{N}$ Inflation?** In finite samples, testing on unseen sample $N + 1$ requires breaking ties conservatively to prevent coverage under-coverage.
- **Deterministic Serialization:** The calibrated threshold $\hat{q}_\alpha$ is tracked as a Git-committed artifact, ensuring online FastAPI pods evaluate identical bounds.
- **Zero Retraining:** Calibration operates entirely on cached embeddings in < 2.5 seconds on CPU.

Distribution-Free Set Construction

- **The Conformal Decision Rule:** For any unseen test observation x_{test}, admit class k if its predicted probability meets the conformal cutoff:

$$\mathcal{C}(x_{\text{test}}) = \{k \in \{1, \ldots, K\} : \hat{\pi}_k(x_{\text{test}}) \geq 1 - \hat{q}_\alpha\}$$

- **Marginal Guarantee:** Under exchangeability:

$$\mathbb{P}\left(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})\right) \geq 1 - \alpha$$

- **Distribution-Free Property:** Holds unconditionally without Gaussian assumptions, linearity, or calibrated logits.

Strict Partition Isolation Invariant

- **Zero Leakage from `val_embeddings.pt`:** The validation split guides Optuna HPO and early stopping. Its residuals are optimistically biased.
- **Untouched `cal_embeddings.pt`:** Calibration split is strictly isolated from model gradient updates and hyperparameter selection.
- **Dynamic Adaptivity:** Clean canonical photos produce $|\mathcal{C}| = 1$; fine-grained sister taxa yield multi-class sets; corruptions yield $\emptyset$.

Thermodynamic Dual Gate

Helmholtz Free Energy & Shannon Entropy Decomposition

Thermodynamic Energy Derivation

Helmholtz Free Energy $E(x; T)$ for logit vector $z \in \mathbb{R}^K$:

$$E(x; T) = -T \ln \sum_{k=1}^{K} \exp\left(\frac{z_k}{T}\right)$$

Substituting Gibbs probabilities $\hat{\pi}_k(x) = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}$:

$$E(x; T) = \underbrace{\sum_{k=1}^{K} \hat{\pi}_k(x)(-z_k)}_{U(x)\text{ (Expected Internal Energy)}} \underbrace{-T \cdot \left(-\sum_{k=1}^{K} \hat{\pi}_k(x) \ln \hat{\pi}_k(x)\right)}_{H(\hat{\pi})\text{ (Shannon Entropy)}}$$

The Softmax Shift-Invariance Flaw

- **Softmax Compression:** $\text{softmax}(z) = \text{softmax}(z + c)$. If all logits are tiny ($z_k \approx -10$), softmax still forces $\sum \hat{\pi}_k = 1$, yielding false confidence on non-biological objects.
- **Energy Magnitude Preserved:** Free energy scales linearly with logit magnitudes:

$$z_k \ll 0 \implies E(x; T) \gg \tau_{\text{OOD}}$$

- **Dual Role:** $U(x)$ measures activation deficit while $H(\hat{\pi})$ measures decision ambiguity.

Predictive Entropy & Triage

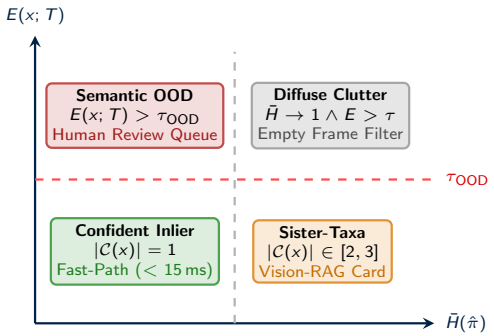
Normalized Shannon Entropy for Clutter & Outlier Rejection

Normalized Entropy Formulation

Normalized Shannon Entropy bounds predictive dispersion to $[0, 1]$:

$$\bar{H}(\hat{\pi}) = -\frac{1}{\ln K} \sum_{k=1}^{K} \hat{\pi}_k(x) \ln \hat{\pi}_k(x) \in [0, 1]$$

- **Empty Camera Traps / Blur:** Wind-blown foliage drives $\hat{\pi} \to \mathcal{U}(1/K)$, so $\bar{H}(\hat{\pi}) \to 1.0$.
- **Margin Sampling Pair:** Evaluates margin $M(x) = \hat{\pi}_{(1)}(x) - \hat{\pi}_{(2)}(x)$ in `active_learning/query.py`.



Pre-Classification Safety Gate

- **The Softmax Trap:** Probabilities normalized over known classes cannot reliably detect when an image does not belong to ANY known biological category (e.g., an indoor living room, vehicle, or macro photograph of rock mineral).
- **Energy-Based Scores:** Computes the negative Helmholtz free energy of the un-normalized logits:

$$E(x; f) = -T \cdot \log \sum_{i=1}^{K} e^{f_i(x)/T}$$

- **Engineering Rationale:** Observations with energy exceeding the calibrated threshold are immediately flagged as OOD prior to classification inference, shielding downstream pipelines from nonsensical data.

Visual Saliency & Interpretability

- **Validation by Taxonomists:** Ecological validation requires verifying that the model relies on genuine diagnostic morphological features (e.g., petal shape or antennae structure) rather than spurious background correlations (e.g., garden fences).
- **Forward-Hook Saliency (Eigen-CAM):** Standard Grad-CAM requires a backward pass that doubles latency. Our implementation utilizes PyTorch forward hooks on penultimate feature activations.
- **Engineering Rationale:** Generates class-discriminative visual saliency heatmaps in under 2 milliseconds without backward gradient overhead, ensuring real-time interpretability within low-latency service constraints.

TaxonVision-MLOps Architecture

CHAPTER 11

CI/CD & Kubernetes Deployment

Declarative Jenkins Pipelines, GHCR Containers & Bare-Metal k3s Orchestration

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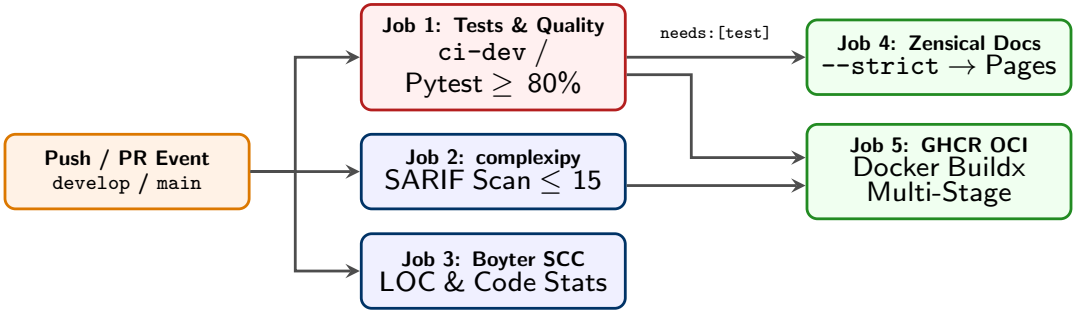
Status: **[Implemented & Verified]**

Security & Network Boundary Architecture

- **Public Cloud Enclave (GitHub Actions):** Pull request validation (Ruff linting, Mypy strict typing, complexipy cognitive complexity ≤ 15, open license audit, OKF validation). Zero access to bare-metal hardware or production secrets.
- **The Bridge:** PR merge to `develop` or `main` $\rightarrow$ Signed commit $\rightarrow$ Cloudflare Zero Trust outbound-authenticated webhook to private Jenkins controller on `hive-mind`.
- **Private On-Premise Enclave (Jenkins & k3s):** Secret injection via Linux Secret Service / unified HMAC tokens. Dedicated GPU training and k3s deployment with zero inbound firewall ports opened.

Cloud Quality Gates

`.github/workflows/ci.yml`: 5-Job Parallel & Sequential DAG



Isolation & Verification Boundary

- **Isolated CPU Runners:** Ubuntu-latest `ci-dev` runner; zero network ingress to bare-metal hardware or production secrets.
- **DagsHub Checkpoints:** Pulls minimal reference model checkpoints via DVC content-addressed cache.
- **Two-Pass Pytest:** Pass 1 enforces branch coverage $\geq 80\%$; Pass 2 verifies integration & conformal guarantees.

Static Analysis & Container Registry

- **Cognitive Complexity:** `complexipy` outputs SARIF, uploaded via `upload-sarif@v4` to GitHub Security tab.
- **Strict Documentation:** Zensical build compiles MkDocs with `--strict`, publishing to GitHub Pages on `main`.
- **GHCR Packaging:** Docker Buildx creates multi-stage container tagged with Git SHA and `latest`.

Cloud Quality Gates

Automated Repository Maintenance Workflows

Scheduled Evidence Indexing (`.github/workflows/openwiki-update.yml`)

- **Trigger & Cadence:** Scheduled weekly cron (`0 8 * * 1`, Monday 08:00 UTC) with manual `workflow_dispatch` fallback.
- **Autonomous Indexer:** Executes Gemini 3.8 Flash via OpenWiki subagent protocol to audit codebase claims against ground-truth unit tests.
- **Incremental Knowledge Bundle:** Updates markdown dossiers, dependency invariants, and claim verification receipts in `openwiki/`.
- **Isolated Commit Bot:** Uses a dedicated GitHub Actions bot token to push signed documentation updates with zero impact on production pipelines.

Semantic Tag Release Pipeline (`.github/workflows/release.yaml`)

- **Semantic Tag Trigger:** Invoked automatically upon pushing annotated Git tags matching `v*.*.*`.
- **Version Invariant Verification:** Strictly asserts version parity between `pyproject.toml` and runtime `taxon_vision.__version__`.
- **Hermetic Wheel Packaging:** Runs `pixi run -e dev flit build` to produce immutable wheels and source distribution tarballs.
- **Cryptographic Release Draft:** Generates SHA256 checksums and drafts an official GitHub Release with automated changelogs.

Six Declarative Pipeline Stages

- **Stage 1 (Checkout & Secrets):** Ephemeral workspace hydration; fetches credentials from Linux Secret Service / DagsHub token.
- **Stage 2 (Environment Hydration):** `pixi install --frozen` with NVMe Rattler cache mount (`/root/.cache/rattler/cache`) avoiding redundant package downloads.
- **Stage 3 (DVC Pull):** Fast download of Parquet manifests and embeddings via `s3://dvc` Content-Addressed Storage.
- **Stage 4 (GPU Training & Optuna):** Strict 5.0 GB VRAM boundary, 150-minute stage timeout (pipeline 240m), `PYTHONUNBUFFERED=1` unbuffered live logging to console.
- **Stage 5 (ONNX Export & INT8 PTQ):** Static spatial $3 \times 224 \times 224$, dynamic batch {`0: "batch_size"`}, QDQ operator insertion.
- **Stage 6 (Artifact Push & Promotion):** `dvc push` of final weights and GHCR OCI container tag promotion.

On-Premise Automation

Jenkinsfile: Parameters, Agent Pod & Secret Bindings

Declarative Pipeline Parameters

- `RETRAIN_MODEL`: Boolean switch (default: `false`); forces execution of Stage 6 GPU retraining and DVC reproduction.
- `TUNE_HYPERPARAMETERS`: Boolean switch (default: `false`); enables Optuna Bayesian search over CBL loss and learning rate.
- `EXTRACTOR`: Choice parameter (`dinov3`, `bioclip2`, `dinov2`, `mobilenetv4`); selects candidate feature backbone.

Ephemeral K8s Agent & Bindings

- **Agent Pod:** Dynamically provisioned in `jenkins-ci` namespace using `ghcr.io/prefix-dev/pixi:latest`.
- **Rattler Cache:** Mounts `emptyDir` volume at `/root/.cache/rattler/cache` for fast Conda solver cache.
- **Credential Ingestion:** Injects `DAGSHUB_TOKEN` from Jenkins credentials store; Linux Secret Service integration.
- **Runtime Governance:** 240m pipeline timeout; `buildDiscarder` keeps 15 builds; `PYTHONUNBUFFERED=1`.

On-Premise Automation

`Jenkinsfile`: 7-Stage Enterprise Execution Flow



Stages 1–5: Fast CPU Gating & Verification

- **Deterministic Environment:** Mounts Rattler package cache; parallelizes Ruff linting, Mypy strict typing, and OKF v0.2 validation.
- **Two-Pass Testing & Docs:** Fast unit suite fails immediately on regression; integration pass verifies conformal bounds; Zensical builds MkDocs.

Stages 6–7: Conditional GPU Training & Rolling Rollout

- **Stage 6 Trigger Invariant:** Activates on `develop`/`main` if changesets touch `data/**`, `models/**`, `pipelines/**`, or `RETRAIN_MODEL=true`. Swaps to GPU pod with 150m stage timeout.
- **Stage 7 Production Release:** Applies `deploy/k8s/api-deployment.yaml` to `taxon-vision` namespace and monitors rollout status.

Ephemeral GPU Runner

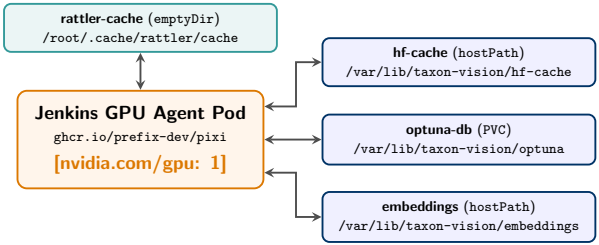
K3s GPU Pod Architecture & Storage Hierarchy

Pod Specification Details

- **Namespace & SA:** Deployed to `jenkins-ci` under dedicated `jenkins-agent` ServiceAccount.
- **Compute Limits:** Requests 2 GiB RAM / 1 CPU; limits 16 GiB RAM, 8 CPUs, and `nvidia.com/gpu: "1"`.
- **VRAM Partitioning:** Strict 5.0 GB ceiling enforced on virtual slice to prevent interfering with production serving.
- **Node Affinity:** Bound to physical host `hive-mind` with direct NVMe volume attachments.

Storage Mounting Architecture

- **Ephemeral Scratch:** `rattler-cache` (`emptyDir`) provides container-isolated Pixi environment caching.
- **Host Direct NVMe:** Direct `hostPath` mounts for Hugging Face weights and DVC tensors avoid redundant downloads.
- **Persistent State:** `optuna-db` via PVC preserves HPO study trials across ephemeral agent teardowns.



Ephemeral GPU Runner

jenkins_gpu_train.sh: 5-Step Execution Lifecycle

1. **Bootstrap & Credentials Ingestion**
   Installs `git` and `kubectl` via Pixi global; validates SeaweedFS S3 credentials (`s3.json` / `taxonadmin`); configures local S3 endpoint `http://local-seaweedfs:8333`.

2. **Embedding Cache Priming (Host & SeaweedFS Tier)**
   Checks `/var/lib/taxon-vision/embeddings/*.pt` on host NVMe or syncs from `s3://dvc/embeddings`; bypasses redundant ViT/DINOv3 backbone extraction for identical manifests.

3. **DVC Pipeline Execution (`dvc repro`)**
   Invokes `dvc repro` with Optuna overrides (`-S training.tune=true`, `-S model.extractor=...`); trains Class-Balanced Loss head; exports INT8 quantized ONNX model.

4. **Multi-Tier Cache Synchronization**
   Synchronizes newly generated `.pt` tensors back to host NVMe tier; pushes intermediate Epistemic Diamond artifacts to in-cluster `local-seaweedfs`.

5. **DagsHub Remote Push & Lineage Lock**
   Pushes production checkpoints (`head.pt`, `model_int8.onnx`) to DagsHub S3 remote (`origin`); auto-commits updated `dvc.lock`, `conformal_config.yaml`, and evaluation reports with `[skip ci]`.

Automated Quality & Complexity Enforcers

- **The "Research Spaghetti" Problem:** ML codebases often decay into deeply nested loops, global states, and untyped tensors.
- **Strict Automated Gates:** TaxonVision enforces code quality through Ruff (formatting/linting) and Mypy (static typing).
- **Cognitive Complexity Limit:** `complexipy` restricts functions to a cognitive complexity of ≤ 15.
- **Refactoring Mandate:** Exceeding the complexity limit fails the pre-commit gate, forcing modular architecture and clean abstraction.

Multi-Stage Container & Deployment Spectrum

- **Multi-Stage Dockerfile Optimization:**
  - **Stage 1 (`builder`):** Ubuntu 24.04 with full Pixi toolchain, compilers, and Rattler build cache.
  - **Stage 2 (`runtime`):** Stripped minimal Ubuntu image copying solely the compiled environment (`/app/.pixi/envs/default`) and application source. Final image size < 1.5 GB (reduced from > 10 GB).
- **Local Development (`flagship`):** `deploy/compose/docker-compose.local.yaml` orchestration with local MinIO, mock DagsHub S3, and Prometheus scraper for zero-cloud offline debugging.
- **Bare-Metal Production (`hive-mind`):** Declarative Kubernetes (k3s) manifests with Traefik ingress controller, zero-downtime rolling updates, and NVIDIA GPU time-slicing daemonsets.

Zero-Downtime Rolling Update Topology

- **The Downtime Challenge:** Updating an inference service must avoid dropping HTTP connections or returning 502 Bad Gateway errors.
- **Health Probes Configuration:** Kubernetes Deployment manifests define readiness and liveness probes targeting the FastAPI `/health` endpoint.
- **Model Warmup Sequence:** Kubernetes waits for model weights to load into GPU memory and warm up before declaring readiness.
- **Seamless Cutover:** Ingress traffic is diverted to new pods only when fully ready, and older replica pods are safely terminated.

Production API Deployment (deploy/k8s/api-deployment.yaml)

- **High-Availability Replicas:** 2 pods with `RollingUpdate` strategy (`maxSurge: 1`, `maxUnavailable: 0`) ensuring zero connection drops during rollouts.
- **Zero-Root Security Context:** Pod runs as non-root user (`runAsUser: 10001`, `runAsGroup: 10001`); mounts read-only volume for ONNX checkpoints (`/app/models/checkpoints`).
- **Dual Health Probes:**
  - **Liveness Probe:** HTTP GET `/health` (initial delay: 15s, period: 20s); restarts stuck containers.
  - **Readiness Probe:** HTTP GET `/health/ready` (initial delay: 5s, period: 5s, failure threshold: 2); prevents traffic ingress until ONNX sessions warm up.
- **Service Exposure:** NodePort service on port 30080 routed through Traefik ingress.

Zero-Trust Namespace Isolation & RBAC (deploy/k8s/jenkins-agent-rbac.yaml)

- **Namespace Segmentation:** Strict perimeter separating `jenkins-ci` (untrusted build workloads) from `taxon-vision` (production inference).
- **ServiceAccount Privileges:** `jenkins-agent` is granted fine-grained verbs (`create`, `delete`, `get`, `list`, `watch`) solely on Pods in `jenkins-ci`.
- **Deployment Rollout Rights:** Constrained to patch/update `deployments` in `taxon-vision`; cannot inspect secrets or alter cluster network policies.

Cluster Topology

In-Cluster Storage, Optuna PVC & GPU Time-Slicing

In-Cluster S3 Edge Storage & Optuna PVC (deploy/k8s/mlops-storage-and-optuna.yaml)

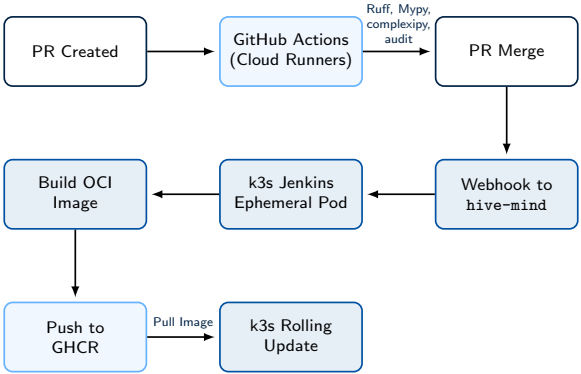
- **SeaweedFS S3 Edge Server:** Lightweight in-cluster S3 daemon on port 8333; configured with `s3.json` for fast local DVC tensor caching.
- **Ultra-Low Latency:** Provides < 2 ms read access for embedding tensors during training, eliminating cloud roundtrip latency.
- **Persistent Volumes:** HostPath-backed Persistent Volume Claims for the shared Optuna SQLite study database (`/var/lib/taxon-vision/optuna`) and raw Parquet manifests.
- **Dual-Remote Architecture:** SeaweedFS acts as ephemeral high-speed local cache; DagsHub serves as canonical cloud cold storage.

NVIDIA GPU Time-Slicing DaemonSet (deploy/k8s/nvidia-time-slicing-config.yaml)

- **Hardware Constraint:** Single physical NVIDIA RTX 5070 Ti (16 GB VRAM) on bare-metal host `hive-mind`.
- **Virtual Partitioning:** NVIDIA Device Plugin ConfigMap splits the physical GPU into 4 virtual slices (`replicas: 4`).
- **Workload Multiplexing:**
  - **Slice 1 & 2:** Production FastAPI inference pods (500 MB model session each).
  - **Slice 3:** Jenkins CI ephemeral training pod (strict 5.0 GB VRAM boundary).
  - **Slice 4:** Interactive research, Jupyter notebooks, or CAM explainability requests.
- **Deadlock Prevention:** Prevents exclusive GPU locking, ensuring CI runs never starve production serving.

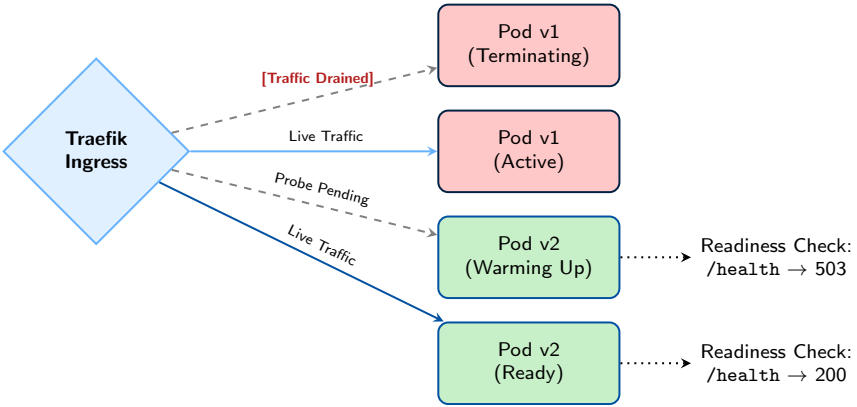
Pipeline Architecture Visualized

Dual CI/CD Lifecycle Diagram



Rolling Update Visualized

Zero-Downtime Traefik Ingress Traffic Switching



CHAPTER 12

Observability & Active Learning

Prometheus Telemetry, Concept Drift Detection & Continuous Human-in-the-Loop Triage

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Status: [Partially Implemented / In Progress]

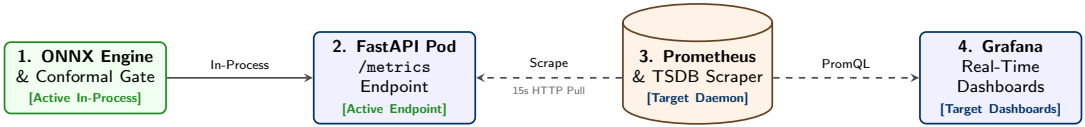
Domain-Specific Telemetry Beyond HTTP 200

- **System Metrics:** p95/p99 inference latency (< 50 ms budget), GPU VRAM utilization **[Target Metric]**.
- **Epistemic Metrics:** Average conformal set size ($|C(X)|$), energy-based OOD anomaly rate **[Target Metric]**.
- **Operational Metrics:** Triage referral rate to human taxonomists (Label Studio queue depth) **[Roadmap]**.
- **Instrumentation Status:** FastAPI `/metrics` endpoint and basic counters implemented in `telemetry.py` **[Implemented]**; external Prometheus scraper and Grafana dashboards **[Target Roadmap]**.

Status: [Target Architecture / Roadmap]

Telemetry Architecture

In-Process Monitoring: FastAPI → Prometheus → Grafana



Latency SLA Histograms

p95 (< 50 ms) & p99 Buckets:
Tracks pre-processing, ONNX inference execution, and conformal set computation latency without locking.

Epistemic & OOD Gauges

Conformal & Energy Gauges:
Real-time prediction set size $|C(x)|$, Helmholtz free energy $E(x)$, and human triage queue referral depth.

Non-Blocking Pull Scrape

15s Scrape Architecture:
FastAPI exposes `/metrics`; Prometheus pulls asynchronously, ensuring zero inference latency overhead.

Production Metric Scrape Architecture

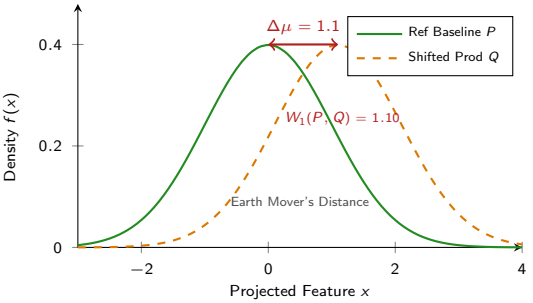
Pull-based collection scraping FastAPI `/metrics` endpoint every 15 seconds.

1. `taxon_predictions_total` **[Implemented in `telemetry.py`]**
2. `taxon_inference_latency_seconds` **[Target Histogram]**
3. `taxon_conformal_set_size` **[Target Gauge]**
4. `taxon_ood_energy` **[Target Distribution]**
5. `taxon_human_referrals_active` **[Roadmap / Target]**

Wasserstein-1 Distance

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| \, dx$$

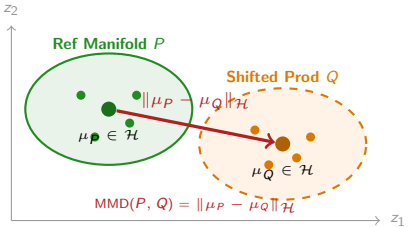
- **Analytical Formulation:** On 1D scalar projections, $W_1(P, Q) = \frac{1}{N} \sum_{i=1}^{N} |x_{(i)} - y_{(i)}|$ after sorting.
- **Sorting Complexity:** $\mathcal{O}(N \log N)$ enables real-time stream evaluation over feature projections.
- **Code Implementation:** Implemented in `src/taxon_vision/monitoring/drift.py` for online batch tracking.
- **Alerting Invariant:** Threshold $\tau_W = 0.15$ triggers triage alerts upon covariate shift from calibration holdout baseline.



Maximum Mean Discrepancy (RKHS MMD)

$$\mathrm{MMD}^2(P, Q) = \mathbb{E}[k(x, x')] - 2\mathbb{E}[k(x, y)] + \mathbb{E}[k(y, y')]$$

- **Kernel Manifold Metric:** Maps embeddings into RKHS $\mathcal{H}$ via Gaussian RBF kernel $k(x, x') = \exp(-\|x - x'\|^2/2\sigma^2)$.
- **Complexity & Cadence:** $\mathcal{O}(N^2 \cdot D)$ quadratic cost over $D = 768/1024$ embeddings; executed as offline scheduled batch job.
- **Evidently AI Integration:** Target framework generating comprehensive visual drift reports, target drift, and feature correlation degradation.
- **Retraining Invariant:** Statistically significant MMD divergence ($\alpha < 0.01$) triggers active learning and DVC retrain pipeline.



The Annotation Bottleneck

Expert taxonomy time is expensive. Randomly sampling from the unlabelled stream yields redundant, low-diversity annotations.

- **BADGE Algorithm:** Batch Active learning by Diverse Gradient Embeddings.
- **Dual Criteria:** It computes hypothetical gradient embeddings representing both *epistemic uncertainty* and *feature diversity*.
- **Selection:** Uses k-MEANS++ clustering in gradient space to select the optimal batch.

Engineering Rationale

By routing only the most mathematically informative samples to Label Studio, we minimise annotation cost while maximising model accuracy improvements per retraining cycle.

Status: **[Partially Implemented / In Progress]**

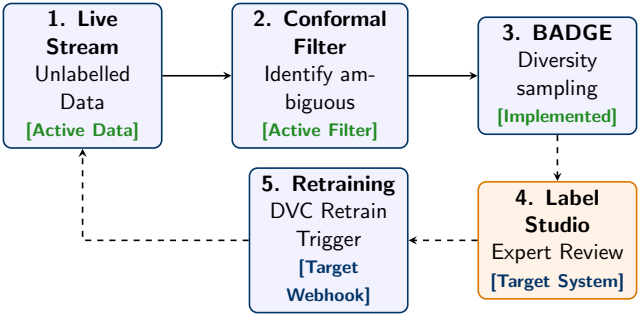
Architectural & Operational Target Blueprint

- **Current Implementation Reality:** Label Studio is **not yet set up or prepared** in our infrastructure. This slide details the target design for our Phase 3 human-in-the-loop triage engine.
- **Triggering Mechanisms:**
  - **Conformal Ambiguity Filter:** Samples where prediction set size $|\hat{C}(x)| > 1$ or empty sets $|\hat{C}(x)| = 0$.
  - **BADGE Diversity Sampling:** Clustering hallucinated gradient embeddings to capture boundary uncertainty **[Algorithm Implemented]**.
- **Annotation Loop (Target):** Automated batch export to Label Studio on DagsHub $\rightarrow$ Domain expert review $\rightarrow$ DVC dataset commit $\rightarrow$ Jenkins retraining webhook **[Roadmap]**.
- **Cost Efficiency:** Prevents sending redundant high-confidence common taxa to human reviewers, cutting labeling expenses by $> 70\%$.

Status: [Target Architecture / Roadmap]

Active Learning Flow

Target Closed-Loop Architecture: Stream → BADGE → Label Studio



CHAPTER 13

Delivery Roadmap & Core Principles

Implementation Milestones, Anti-Pattern Rectification & Production Systems Engineering

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Status: **[Implemented & Verified]**

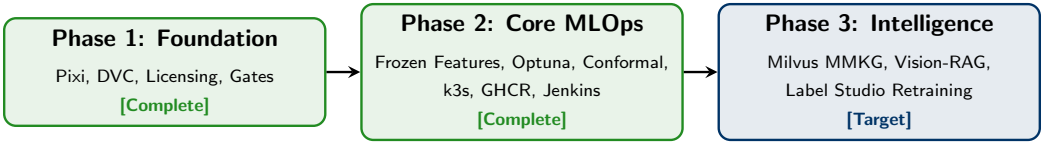
Project Roadmap

Implementation Status vs Architectural Roadmap

TaxonVision Delivery Phases

An honest, transparent breakdown of the current implementation status versus future architectural milestones.

- **Phase 1 (Foundation):** Complete. Pixi toolchain, DVC/S3 streaming, pre-commit gates, license compliance.
- **Phase 2 (Model & MLOps):** Complete. Frozen feature extraction, Class-Balanced Loss, Optuna, Conformal safety, k3s, GHCR, Jenkins.
- **Phase 3 (Continuous Intelligence):** Target. Full Milvus MMKG, Vision-RAG, Label Studio retraining trigger.



Status: [Partially Implemented / In Progress]

Bootcamp vs. Enterprise

Core pedagogical lessons for transitioning into MLOps, contrasting common junior anti-patterns with enterprise architectural solutions.

Monolithic Notebook Sprawl Unversioned, stateful Jupyter notebooks	**Modular Python Packages** Strict complexity limits, typed functions
Local Dataset Hoarding Downloading terabytes to local SSDs	**Zero-Copy S3 Streaming** DVC pointers, Apache Arrow Parquet
Uncalibrated Softmax Trusting raw model probabilities blindly	**Split Conformal Prediction** Mathematically guaranteed safety bounds
Unbudgeted Cloud GPUs Renting unmanaged A100s indefinitely	**Bare-Metal GPU Time-Slicing** Strict memory ceilings on RTX 5070 Ti

Promotion Lifecycle

Study Run → Candidate Model → Automated Conformal Gate → Staging Tag → Production Promotion

Audit Integrity

- **Cryptographic Binding:** Strict cryptographic binding connecting production model version to immutable DVC commit and signed Git release tag.
- **Safety Guarantee:** Prevents deployment of uncalibrated or drifting models.

Enterprise Level 4 Verification Matrix

- **Data Lineage:** 100% deterministic Parquet metadata with DVC SHA256 immutability.
- **Hermetic Toolchains:** Zero reliance on unpinned pip/conda environments; strict lockfiles via Pixi.
- **Mathematical Safety:** Automated test gates asserting conformal coverage $1 - \alpha$ on holdout splits.
- **Continuous Deployment & Observability:** Jenkins OCI image builds and k3s manifests established; streaming drift monitors and Label Studio triage scheduled as Phase 3 roadmap.

Status: **[Partially Implemented / In Progress]**

CHAPTER 14

Multimodal Knowledge Graph & Vision-RAG

Epistemic Ambiguity Resolution, Filtered Vector Retrieval & Clinical-Grade Diagnosis Cards

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Status: [Target Architecture / Roadmap]

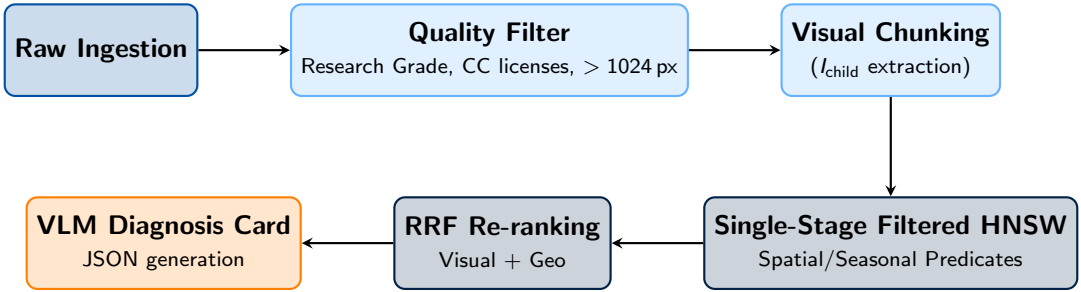
Conformal-Driven Lazy Invocation Policy

- **Throughput Preservation:** Querying dense graph and vector databases on routine inferences exhausts latency budgets.
- **Fast-Path Resolution:** Unambiguous singleton predictions ($|\hat{C}(X)| = 1$) resolve in ≤ 15 ms via INT8 ONNX models.
- **Lazy Activation Gate:** The MMKG pipeline triggers exclusively when the Split Conformal Gate detects epistemic uncertainty.
- **Epistemic Triggers:** Severe classification ambiguity ($|\hat{C}(X)| > 3$) between sibling taxa or complete OOD outliers ($|\hat{C}(X)| = 0$).

Status: [Target Architecture / Roadmap]

MMKG Pipeline Architecture

Ingestion & Retrieval Data Flow

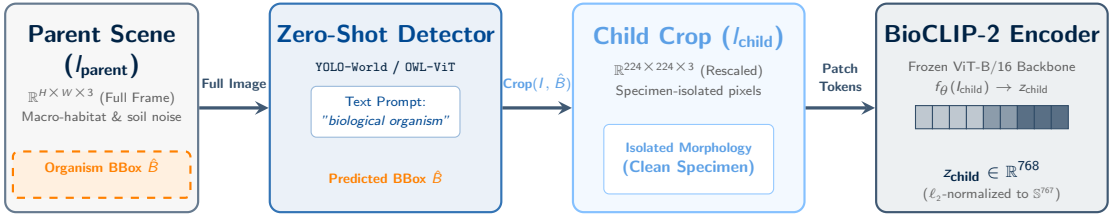


Background Clutter Isolation Strategy

- **The Foliage Noise Problem:** In uncropped citizen-science photos, specimens occupy $< 10\%$ of pixels; soil and sky corrupt embeddings.
- **Zero-Shot Organism Localization:** Upstream zero-shot detectors (YOLO-World / OWL-ViT) isolate primary organism bounding boxes.
- **Focused Child Embedding:** Extracts noise-free BioCLIP-2 embeddings ($z_{\text{child}} \in \mathbb{R}^{768}$) from the isolated specimen crop (I_{child}).
- **Context Preservation:** Uncropped parent scene context (I_{parent}) is retained for macro-habitat and environmental verification.

Visual Pre-Processing

Parent-Child Visual Chunking Diagram



Spatial Noise & Confounder Filtering

- **Environmental Disentanglement:** Discards non-diagnostic background (soil, foliage, human fingers, specimen pins).
- **Receptive Field Maximization:** Rescales the target organism to full 224×224 patch resolution.
- **Zero-Shot Localization:** Open-vocabulary detector handles unannotated taxa without category retraining.

Downstream Embedding Invariants

- **Morphology Isolation:** Precludes ViT attention heads from latching onto spurious background shortcuts.
- **Unit Hypersphere Projection:** Feature vector is ℓ_2-normalized to $\mathbb{S}^{767}$ for optimal cosine retrieval.
- **Context Decoupling:** Macro-habitat context can be independently routed as an auxiliary tabular feature.

Pruning Geographically Impossible Candidates

- **Convergent Evolution Trap:** Unrelated species on distant continents evolve near-identical morphologies (e.g., mimicry).
- **Post-Filtering Failure:** Top-k visual search post-filtering causes candidate starvation when top neighbors are from the wrong continent.
- **Native Single-Stage HNSW Walk:** Spatial predicates (Haversine geodesic $R_{\text{max}} \leq 500$ km) evaluated directly inside C++ graph traversal.
- **Temporal Phenology Bounds:** Seasonal observation month filters ($\Delta_{\text{season}} = 1$) prune phenologically impossible taxa.

Synthesizing Visual Similarity & Ecological Priors

- **Rank Combination:** Combines dense visual cosine similarity ($w_{\text{vis}} = 0.65$) with GBIF historical occurrence density ($w_{\text{geo}} = 0.35$):

$$\text{RRF}(d) = \frac{w_{\text{vis}}}{k + r_{\text{vis}}(d)} + \frac{w_{\text{geo}}}{k + r_{\text{geo}}(d)}$$

- **Distribution Invariance:** Non-linear rank fusion prevents score scale mismatch between dissimilar probability spaces.
- **Robustness:** Constant $k = 60$ dampens top-rank outliers, tilting decisions toward true local ecology.

Grammar-Constrained VLM Reasoning for Triage

- **Taxonomist Context Synthesis:** When ambiguous observations hit human review, query crops and top-3 exemplar specimens are injected into a local VLM (Qwen2-VL-2B).
- **Grammar-Based Sampling:** Strict JSON grammar constraint prevents hallucination and guarantees valid schema adherence.
- **Differential Diagnosis Card:** Emits structured anatomical discriminators, likelihoods, and CAM activation overlays.
- **Label Studio Integration:** Direct rendering in review UI turns triage into rapid expert verification.

Handling Dynamic Biological Taxonomy

- **Dynamic Taxa Evolution:** Biological committees frequently revise classifications: merging species (lumps) or subdividing them (splits).
- **Decoupled Architecture:** Embeddings reside in HNSW vector storage, while taxonomic relationships and metadata reside in DuckDB.
- $\mathcal{O}(1)$ **Taxonomic Lumps:** Species mergers execute as instant relational table updates without touching model weights or embeddings.
- **Audited Taxonomic Splits:** Historical nodes are marked `requires_reannotation = true` and safely queued for expert re-labeling.

The Pillars of Production Systems Engineering

TaxonVision is built on three uncompromising principles that define enterprise MLOps.

- **Reproducibility by Default:** Using Pixi environments, DVC data lineage, and declarative k3s manifests ensures every run is perfectly replicable.
- **Mathematical Safety under Uncertainty:** Split Conformal Prediction guarantees error bounds, refusing confident misclassification.
- **Hardware-Conscious Sustainability:** Bare-metal GPU time-slicing, INT8 ONNX quantization, and zero-allocation inference maximize throughput per watt.